Galactic Astrophysics

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About Me

- PhD in Astrophysics then astrophysics research and teaching
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- DAIM – Centre for Data Science, Artificial Intelligence and Modelling
- MSc conversion course:
- https://www.hull.ac.uk/study/postgraduate/taught/artificial-intelligence-and-data-science-msc

Plan For Today

Morning:

Galaxy rotation curves and dark matter

Afternoon:

Galaxy clusters

Galaxy morphology

Astrophysics of Galaxies: Rotation Curves and Dark Matter

What is a galaxy?

Collections of
- Stars
- Gas
- Dust

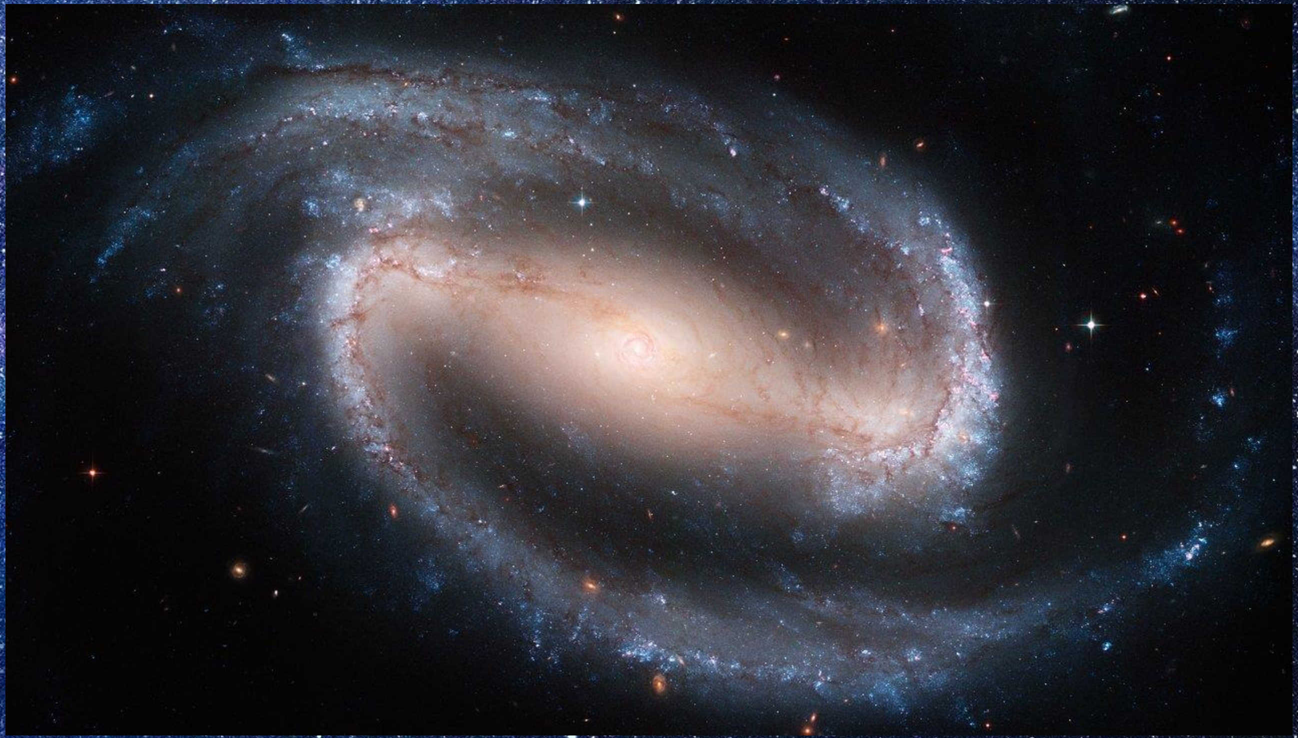
All held together by gravity

Found in different shapes and sizes.

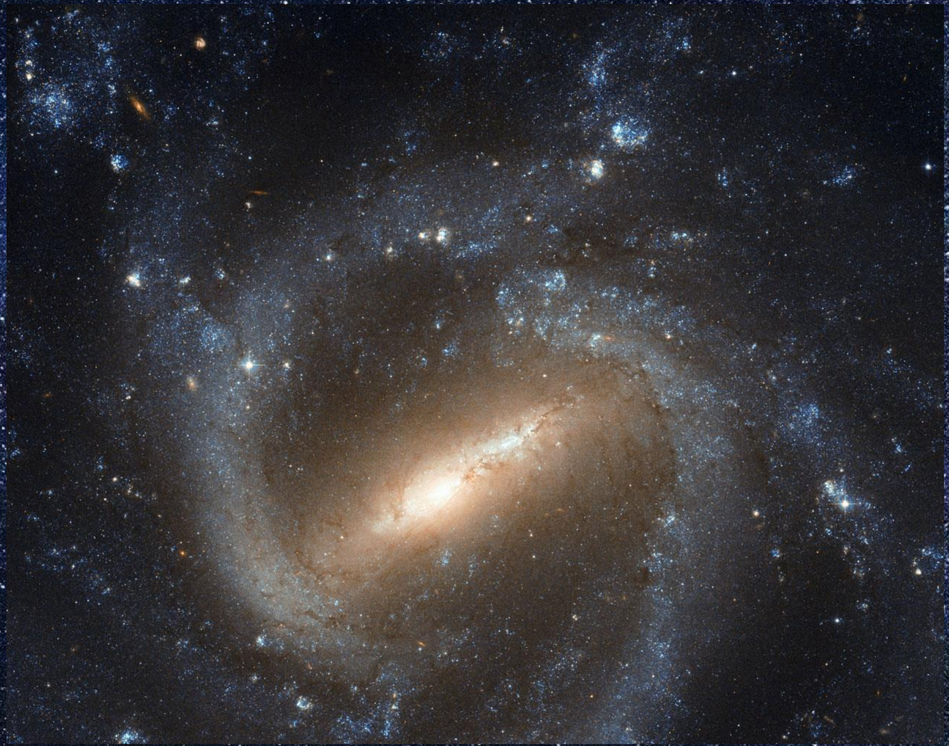
This is a spiral galaxy

Pinwheel galaxy Credit: European Space agency & NASA

Barred Spiral Galaxies

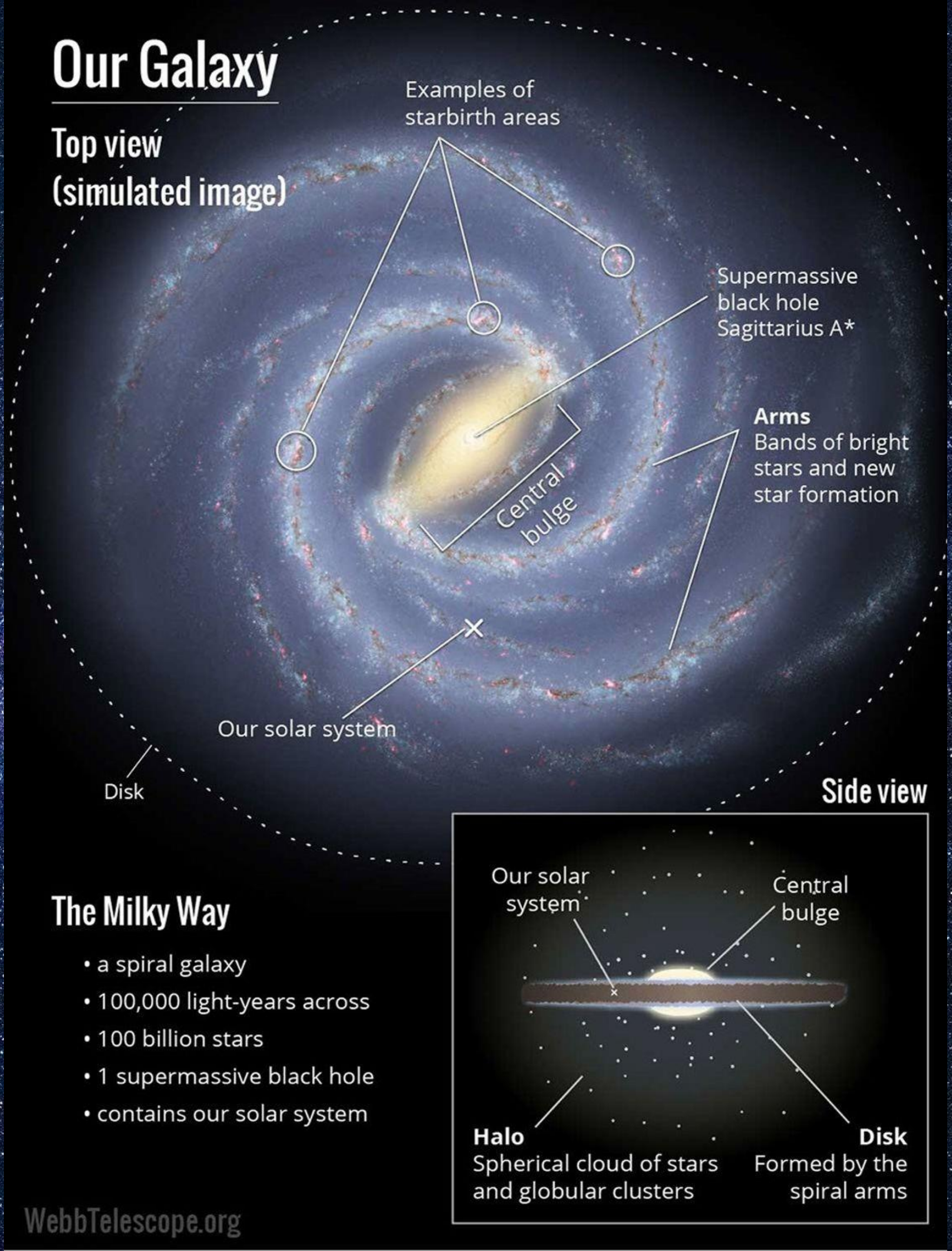


NGC 1300



NGC 1073

Credit: NASA, ESA and the Hubble Heritage Team

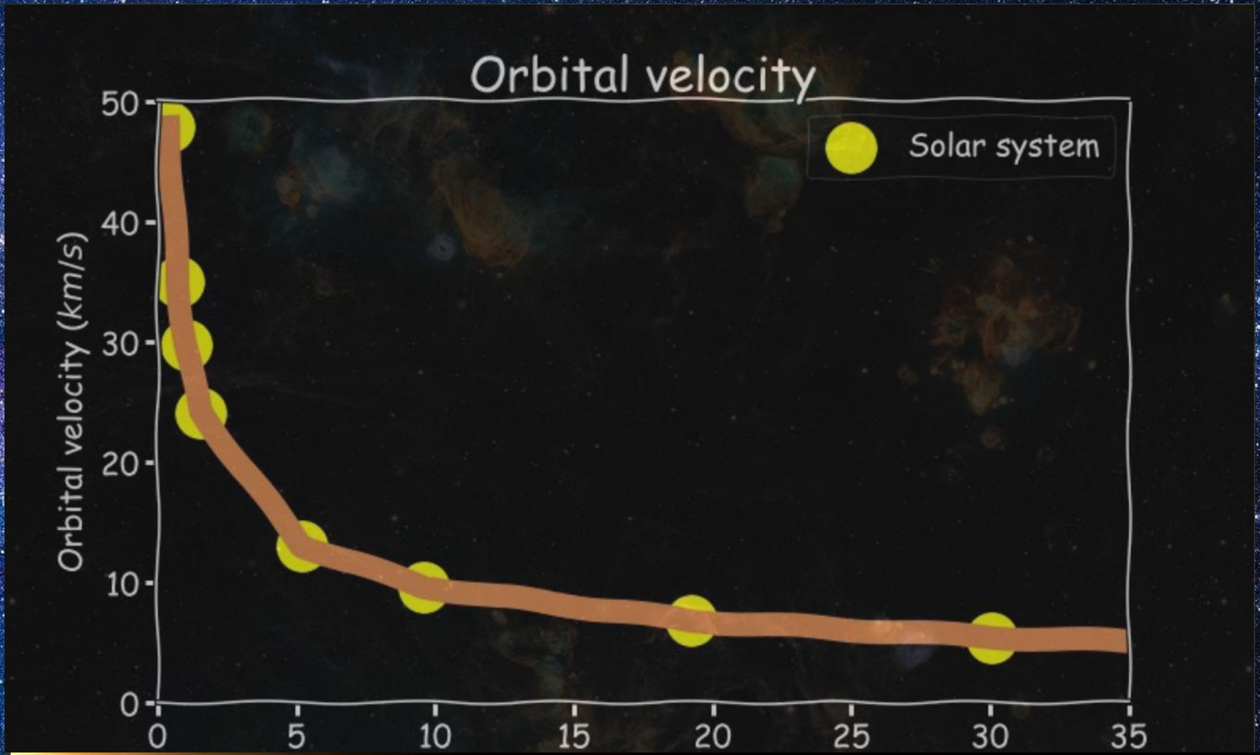


Our Galaxy
Top view
(simulated image)
Examples of starbirth areas
Supermassive black hole Sagittarius A*
Arms
Bands of bright stars and new star formation
Central bulge
Our solar system
Disk
Side view
Our solar system
Central bulge
Halo
Spherical cloud of stars and globular clusters
Disk
Formed by the spiral arms
The Milky Way
• a spiral galaxy
• 100,000 light-years across
• 100 billion stars
• 1 supermassive black hole
• contains our solar system
WebbTelescope.org

Rotation in galaxies

- Spiral galaxies spin like a plate or pizza, but different parts spin with different velocities.
- The bright areas are stars and gas which can be observed directly with telescopes
- Mass is spread out over a large area, very dense in the core, less dense further and further out.

Rotation curves – the solar system



Orbital velocity as a function of distance from centre.

Sun as central mass

What is a rotation curve?

Gravity is an attractive force which acts between all objects that have mass

The size of the force of gravity depends on

- The mass of each object
- The distance between the objects

The gravitational force keeps objects in orbit and determines how fast they move in their orbit.

What is a rotation curve?

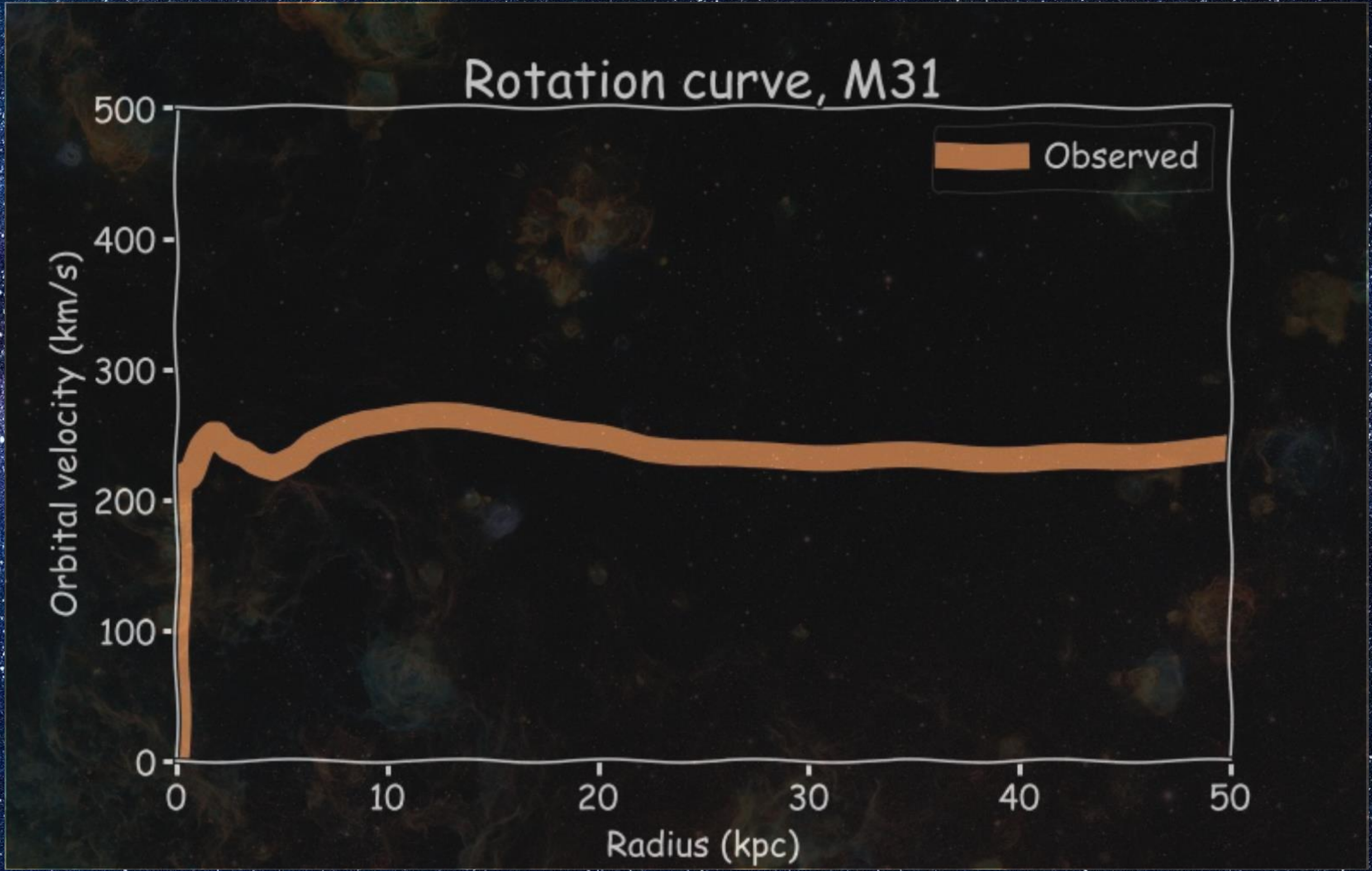
Orbital velocity as a function of distance from centre.

- M is the mass of the Sun
- R is the distance from the Sun
- G is the gravitational constant
- V is the speed of the planet

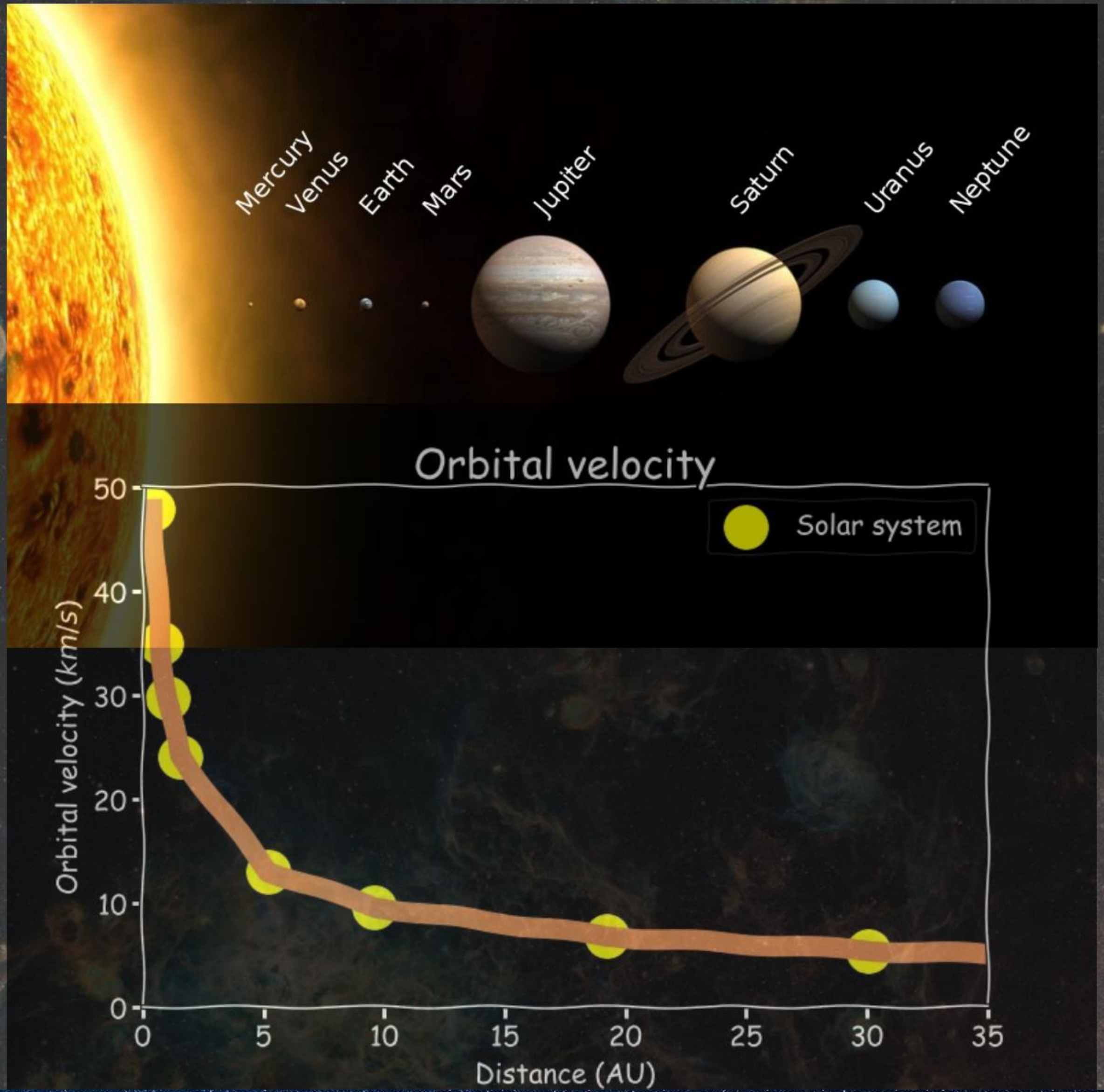
$$v^2 = \frac{GM}{r}$$

- Speed falls of with distance ($\sqrt{\frac{1}{r}}$)

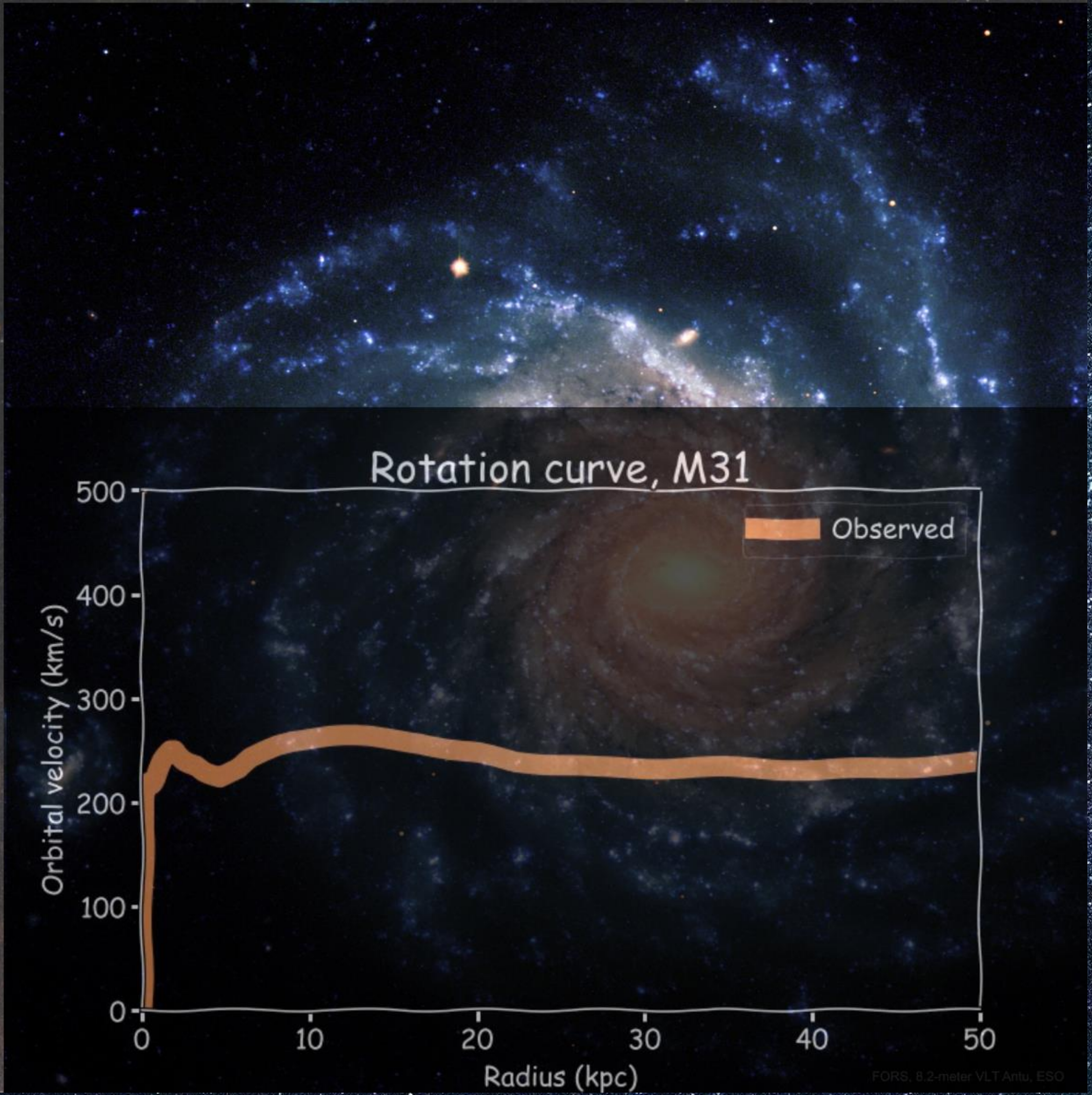
Rotation curves of galaxies



How is this different to a rotation curve for the solar system?



Mercury
Venus
Earth
Mars
Jupiter
Saturn
Uranus
Neptune
Orbital velocity
Solar system
Orbital velocity (km/s)
0
10
20
30
40
50
0
5
10
15
20
25
30
35
Distance (AU)

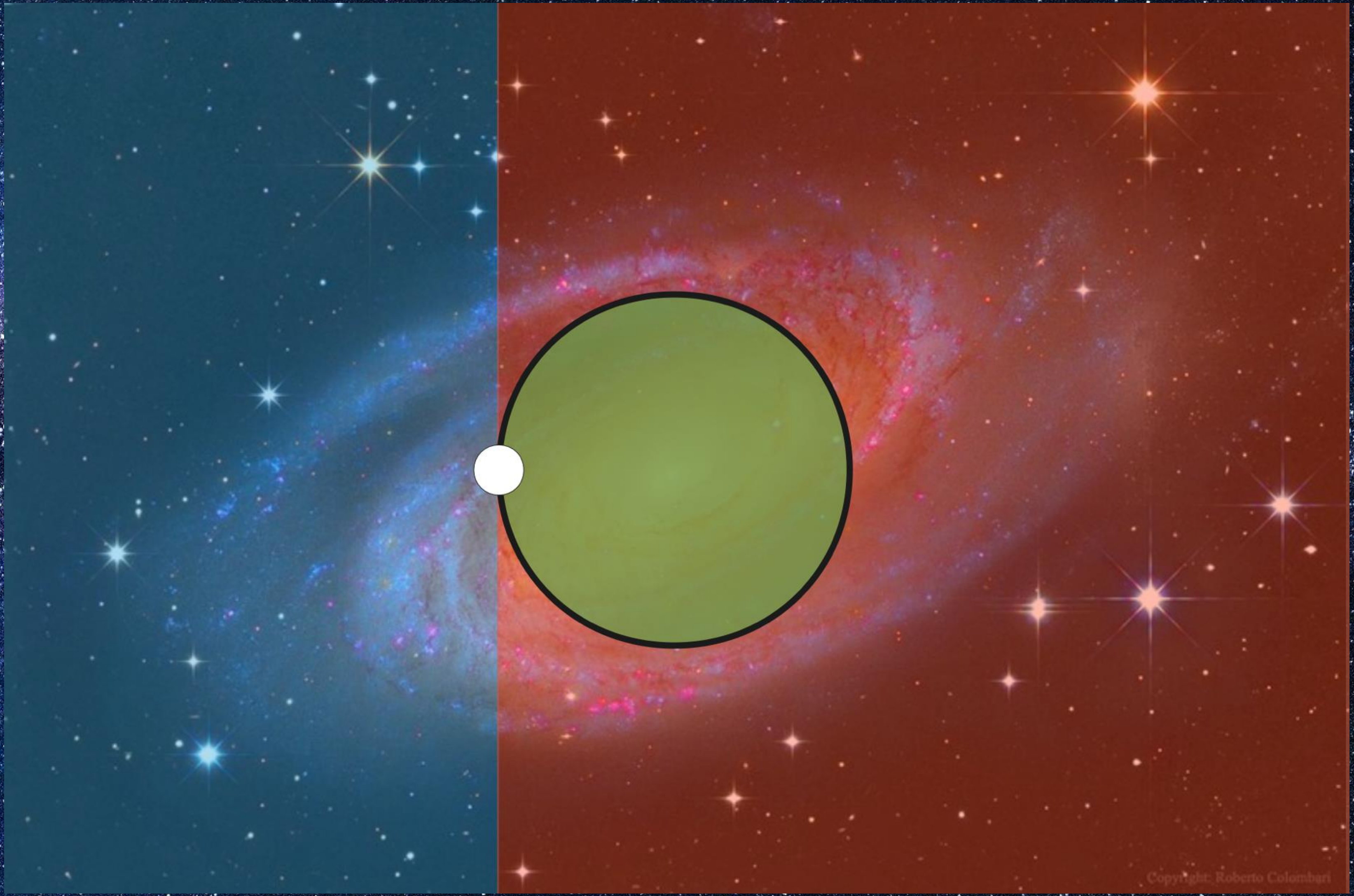


Rotation curve, M31
Observed
Orbital velocity (km/s)
0
100
200
300
400
500
0
10
20
30
40
50
Radius (kpc)
FORS, 8.2-meter VLT Antu, ESO

Rotation Curves of Galaxies

- The solar system had a single mass (the Sun or a star) at the centre.
- A galaxy is spread out over a large area. This raises questions:
  - The force that decides the orbit is the gravitational force. How does the extended mass distribution affect this?
  - What mass should we use in our equation? Can we even use the same equation?

The answer is surprisingly simple



$$V = \sqrt{\frac{GM(r)}{r}}$$

How to measure mass in galaxies

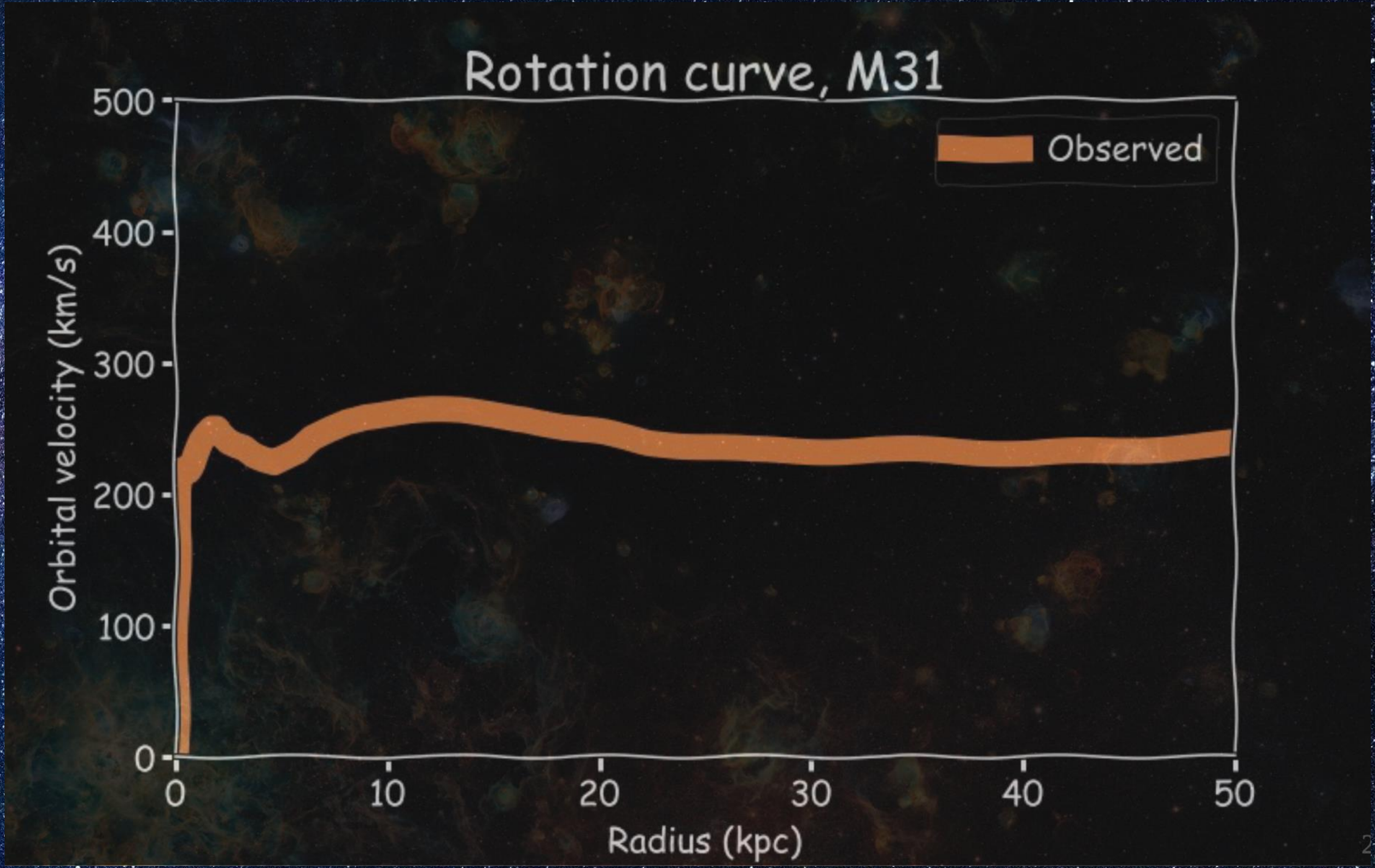
Astronomers can't measure mass directly... We have two methods:

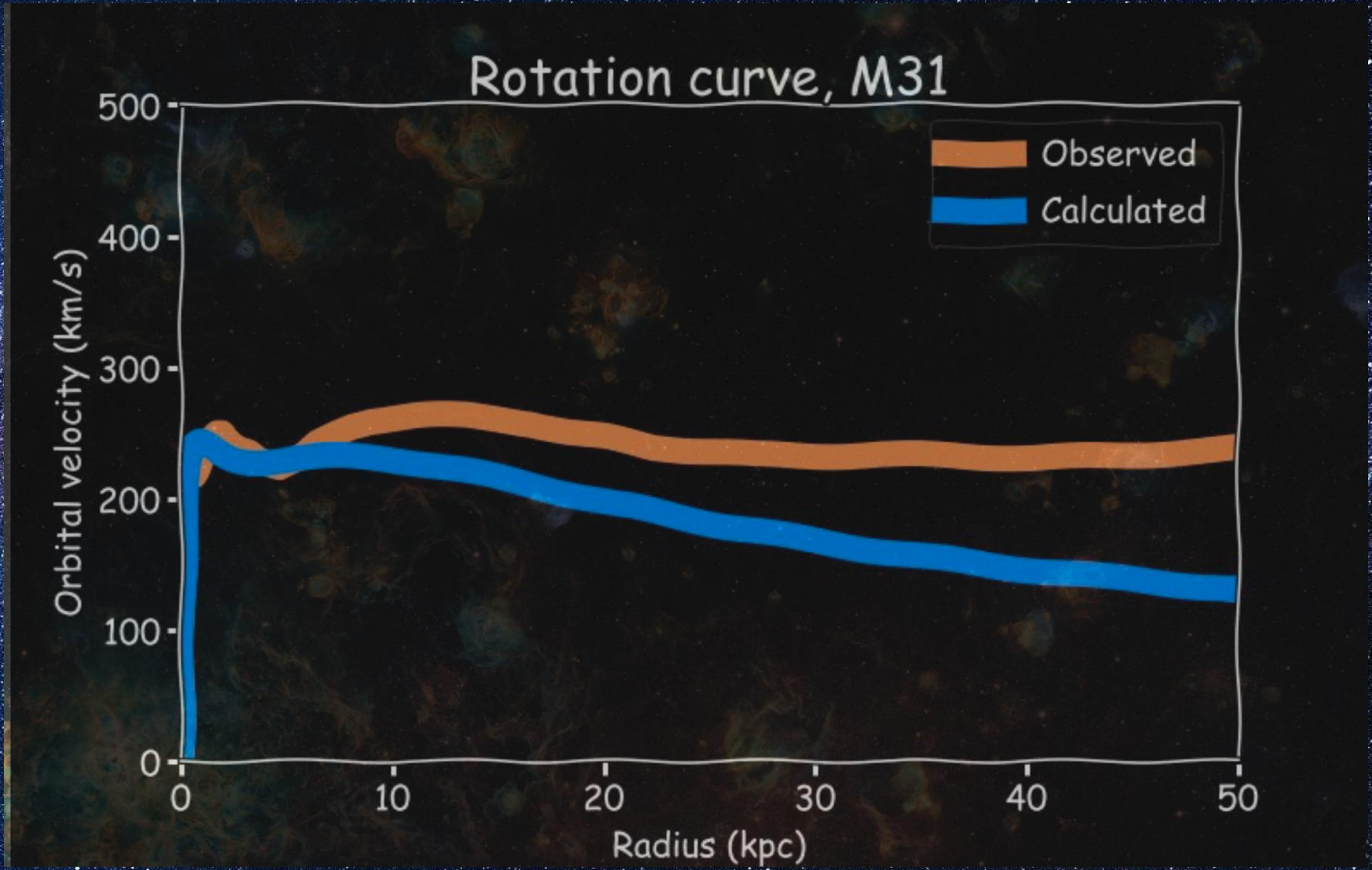
1. Measuring the orbital velocity of stars and gas and use the orbital velocity equation to find the mass

$$V = \sqrt{\frac{GM(r)}{r}}$$

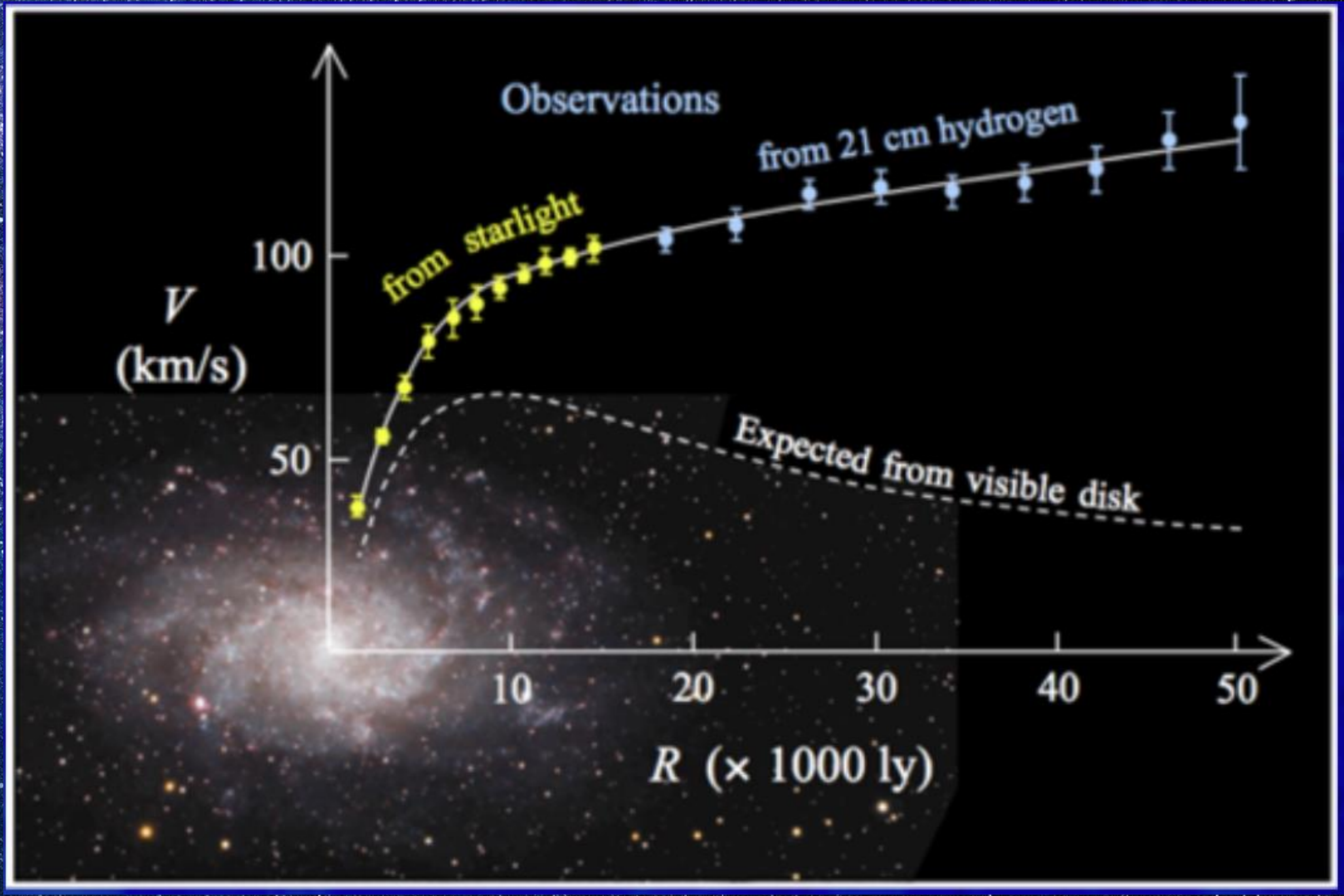
2. Measuring the light of the galaxy, which consists of only of light from stars and gas and since we know how bright one star is, we can infer the total mass (Mass-luminosity relation)

$$M \propto L^{\alpha}$$


Rotation curve, M31
Observed
Orbital velocity (km/s)
500
400
300
200
100
0
0
10
20
30
40
50
Radius (kpc)


Rotation curve, M31
Observed
Calculated
Orbital velocity (km/s)
Radius (kpc)
0
100
200
300
400
500
0
10
20
30
40
50

Where is the missing mass?



Stars on the edge of the Andromeda galaxy are moving as fast as the stars at the centre!

Optical data from Rubin and Ford (1970) plus outer 21 cm data

What is dark matter

- We are made of ‘baryonic’ matter
- Refer to dark matter as ‘non-baryonic’ matter

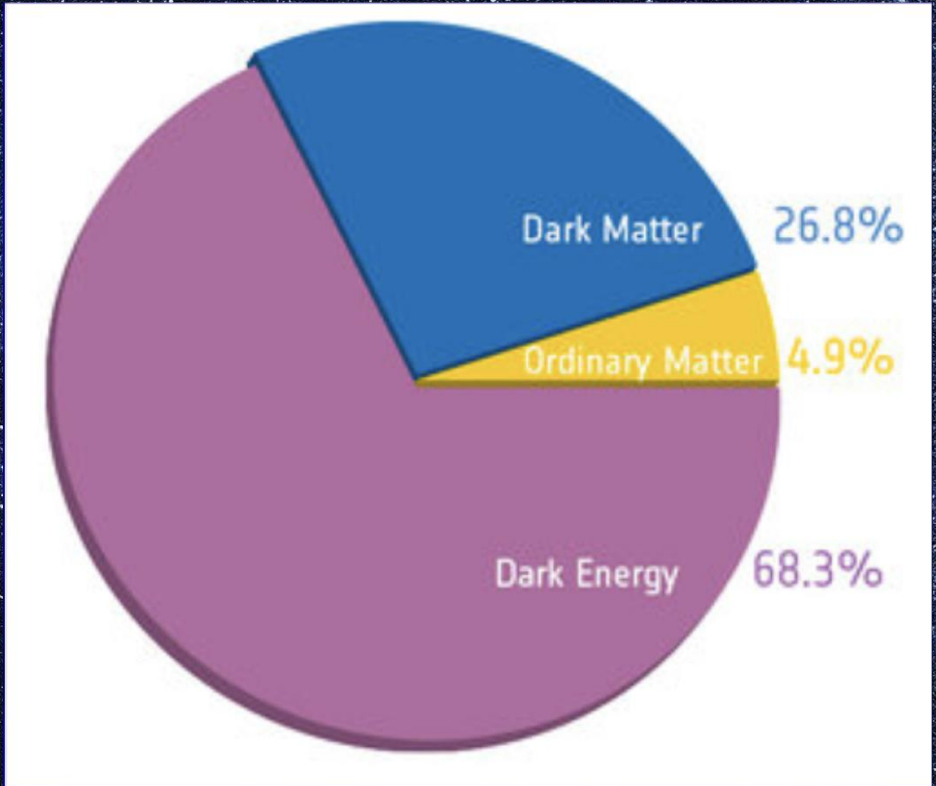
- Both take up space and hold mass

- Dark matter interacts with baryonic matter through gravity
- Dark matter does not interact with light

Lambda-CDM

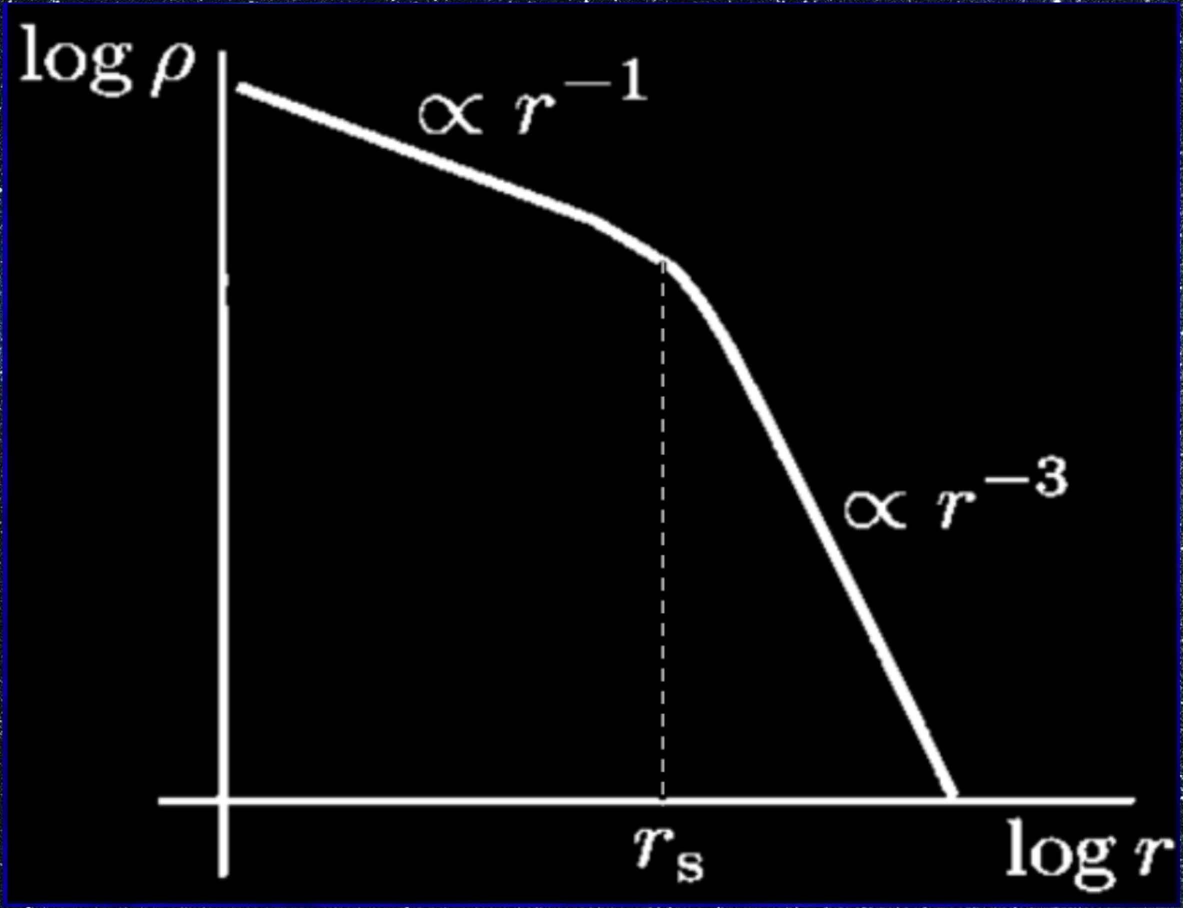
- Cold dark matter is now part of the *Standard Model of Cosmology* in addition to Dark Energy.

- Everything we can see around us and everything we can see in space makes up just 4.9% of the Universe!



Dark matter Mass

- N-body simulations study dark matter structure in CDM cosmology
- Distribution of dark matter follows the distribution of galaxies
- Find a universal density profile fits all DM structures (halos)



Navarro - Frenk - White Profile (1997)

We can describe this distribution with the following density and mass profiles:

$$\rho(r) = \frac{\rho_0}{\frac{r}{r_s}\left(1 + \frac{r}{r_s}\right)^2}$$

$$M = \int_0^{r_{max}} 4\pi r^2 \rho(r) dr$$

r is the radius
r_s is the scale radius

$\rho(r)$ is the density at a radius, r
ρ_0 *is the characteristic density*

ρ_0 and r_s are parameters which vary from halo to halo

Other evidence for dark matter

Gravitational Lensing

- Stars and gas in galaxy clusters only make up about 5% of the total mass
- 80% of the mass in a cluster is in the form of dark matter,
- This mass provides enough gravity to bend the path of light passing by which magnifies galaxies that would ordinarily be too faint for us to see.
- Lensing lets astronomers map where the mass in a galaxy cluster is located.

We will talk about the remaining 15% later!

Credit: NASA, ESA, and J. Lotz and the HFF Team (STScl)

The Bullet Cluster

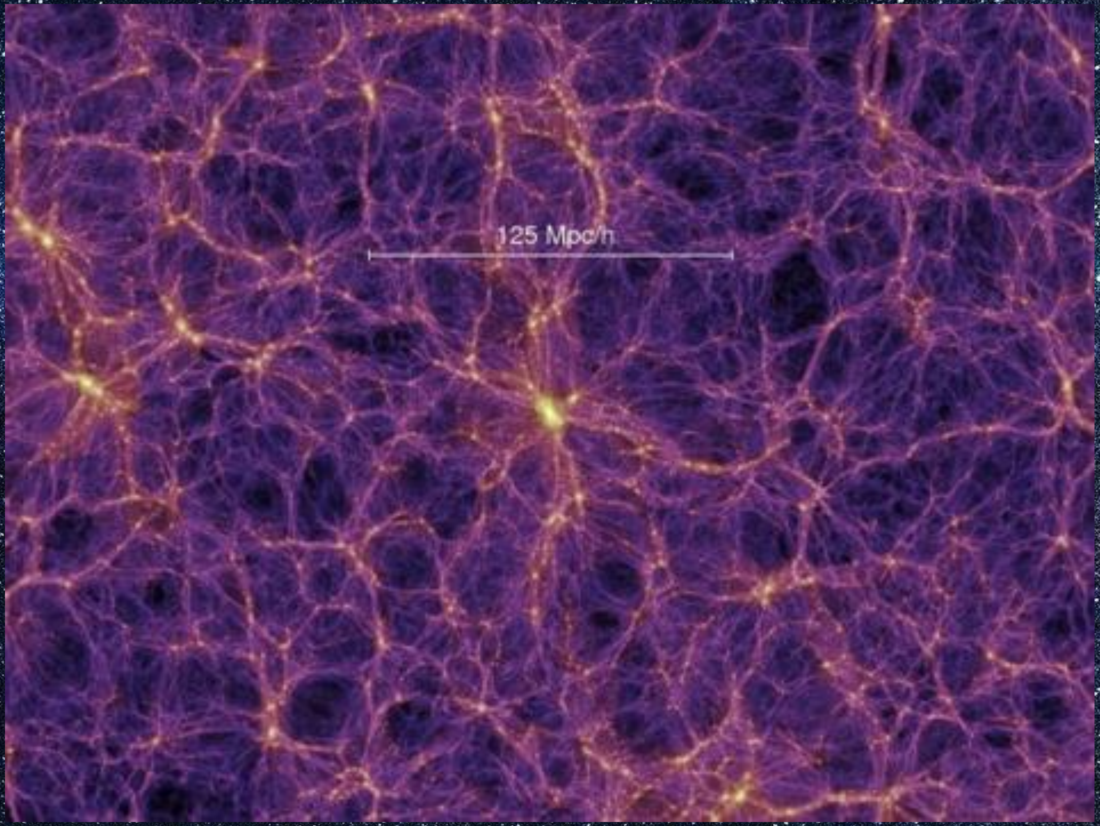
- Collision of two large clusters of galaxies
- Hot gas detected by Chandra in X-rays shows two pink clumps – normal matter
- Blue shows where most of the mass is (determined using gravitational lensing)
- Most of the matter in the clusters (blue) is separate from the normal matter (pink)

End of part 1!

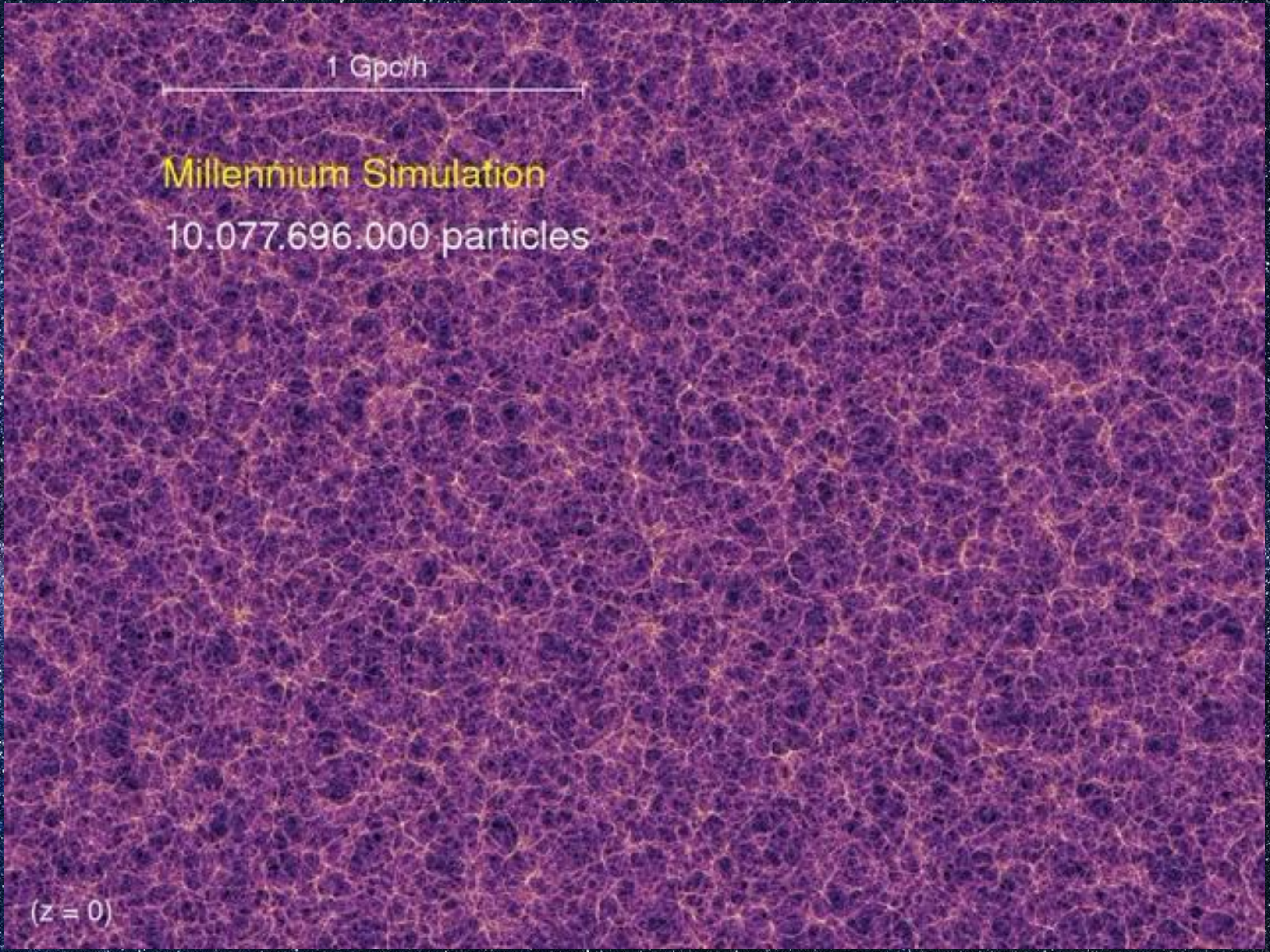
Astrophysics of Galaxies: Large scale structure

Structure of the Universe

- The universe is not uniform
- Matter clumps together
- Concentrated clumps containing galaxies
- Large empty areas in-between called “voids”



Credit: The Millennium Simulation project.


1 Gpc/h
Millennium Simulation
10.077.696.000 particles
(z = 0)

The Environment

Field:

- Regions containing very few (if any) galaxies
- Galaxies are gravitationally alone

Groups:

- Up to ~100 galaxies

Clusters:

- Very crowded, can contain 1000s of galaxies

Roughly 80% of all galaxies located within 5Mpc of the MW are in groups or clusters

Galaxy Clusters

- Most massive, gravitationally bound systems in the universe.
- Can contain 1000s of galaxies.
- Ideal for studying large populations of galaxies.
- Tracer of galaxy evolution.
- Most of the baryonic mass is in the intracluster medium (ICM)

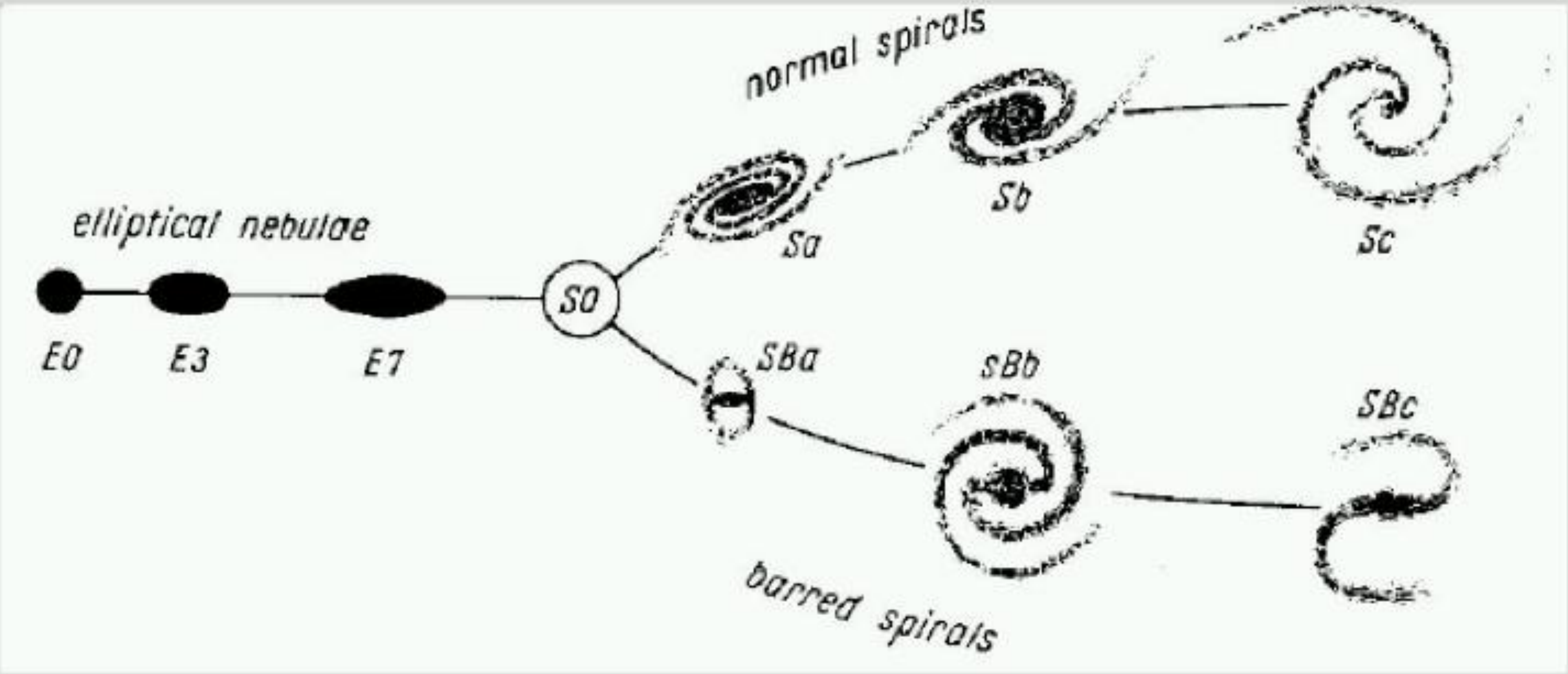
The Coma Cluster

- Largest and most densely populated cluster known
- 321 million light years away
- Mostly elliptical galaxies

Credit: Justin Ng

Morphological classification of galaxies

Edwin Hubble devised a scheme for classifying galaxies, based on their appearance, in his 1936 book *The Realm of the Nebulae*.



Elliptical galaxies **Spiral galaxies**

Galaxy Transformations

Spiral

Time

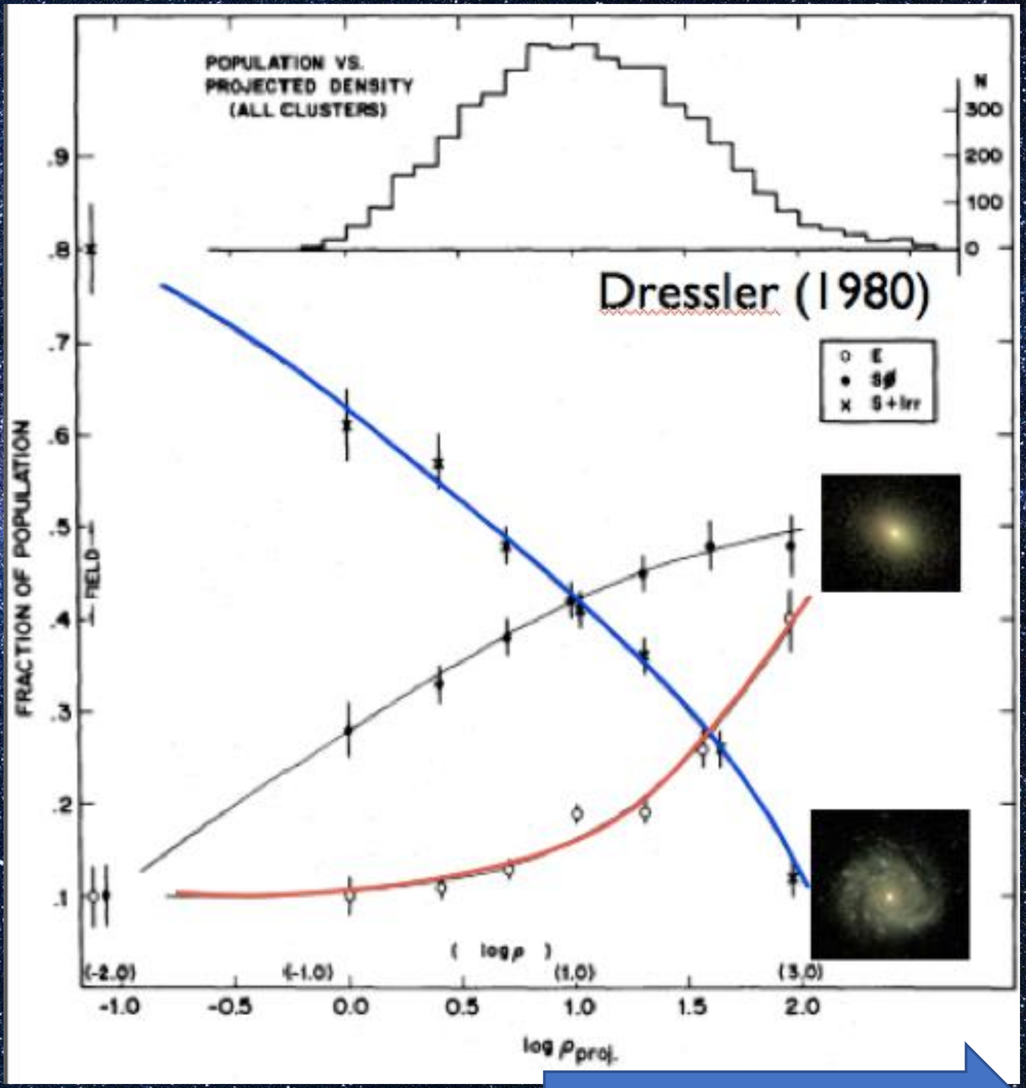
Environment

Elliptical

Around 75% of large galaxies in the field are spirals

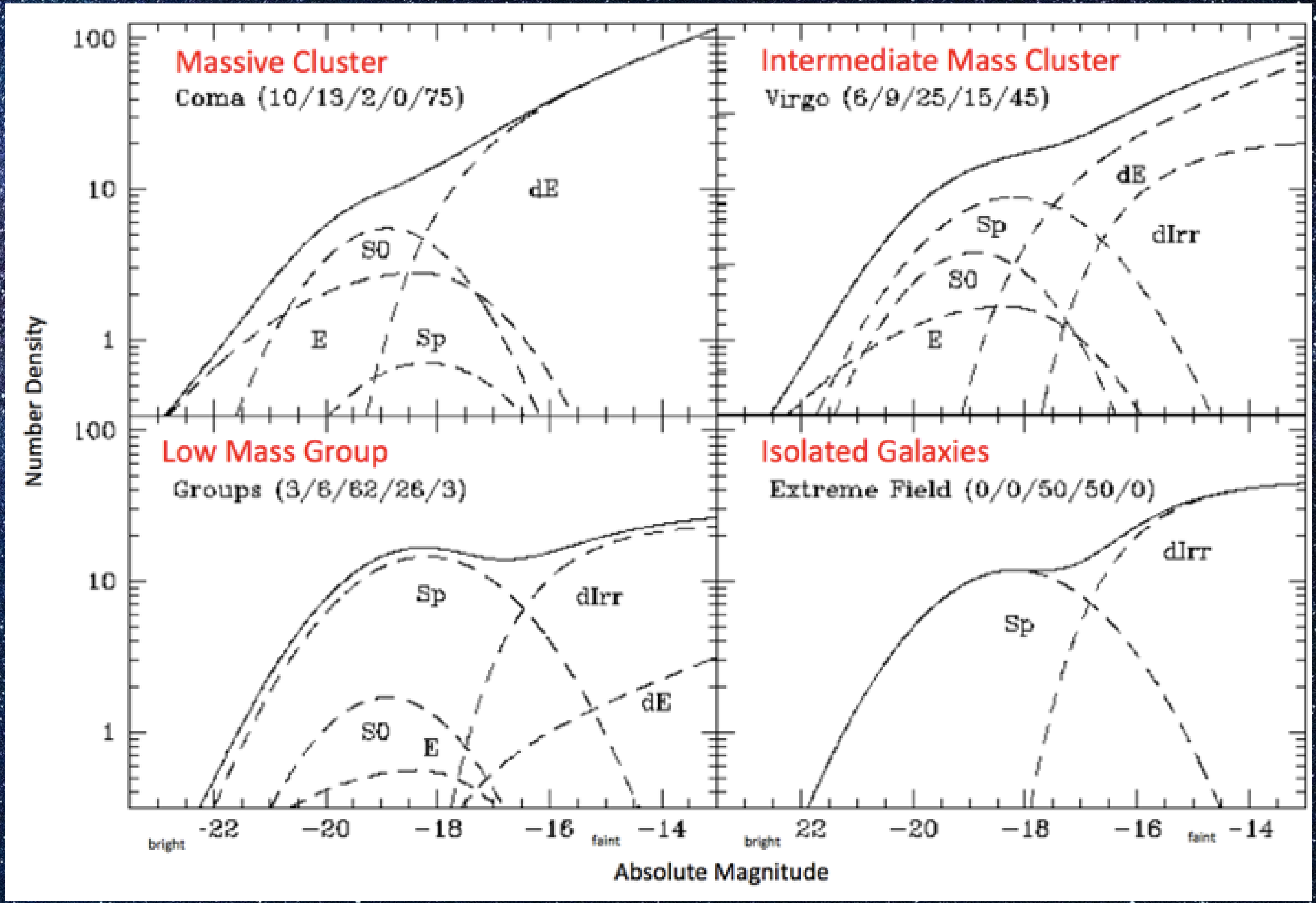
Morphology - Density Relation

Increasing number of galaxies



Higher density – more neighbours

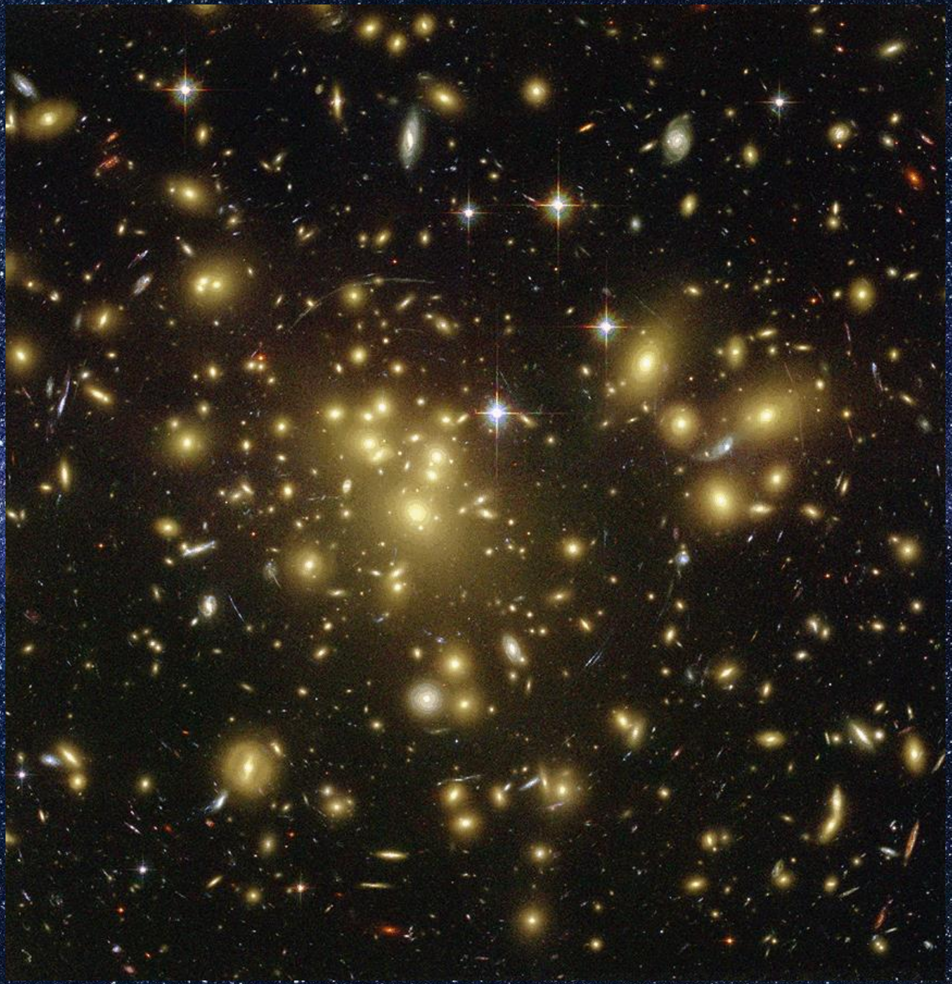
Fraction of galaxies in different environments



Environmental processes

- **Gas starvation/strangulation** - removal of the hot gas halo once the galaxy falls into the cluster.
- **Mergers** - strong internal dynamical response.
- **Tidal stripping -** removal of gas and stars through gravitational force
- **Ram pressure stripping** – removal of gas, indirect influence on morphology.

Intracluster Medium (ICM)



Abell 1689 in optical and X-ray emission
Image courtesy of Optical: NASA/STScI. X-ray: NASA/CXC/MIT/E.-H Peng et al.

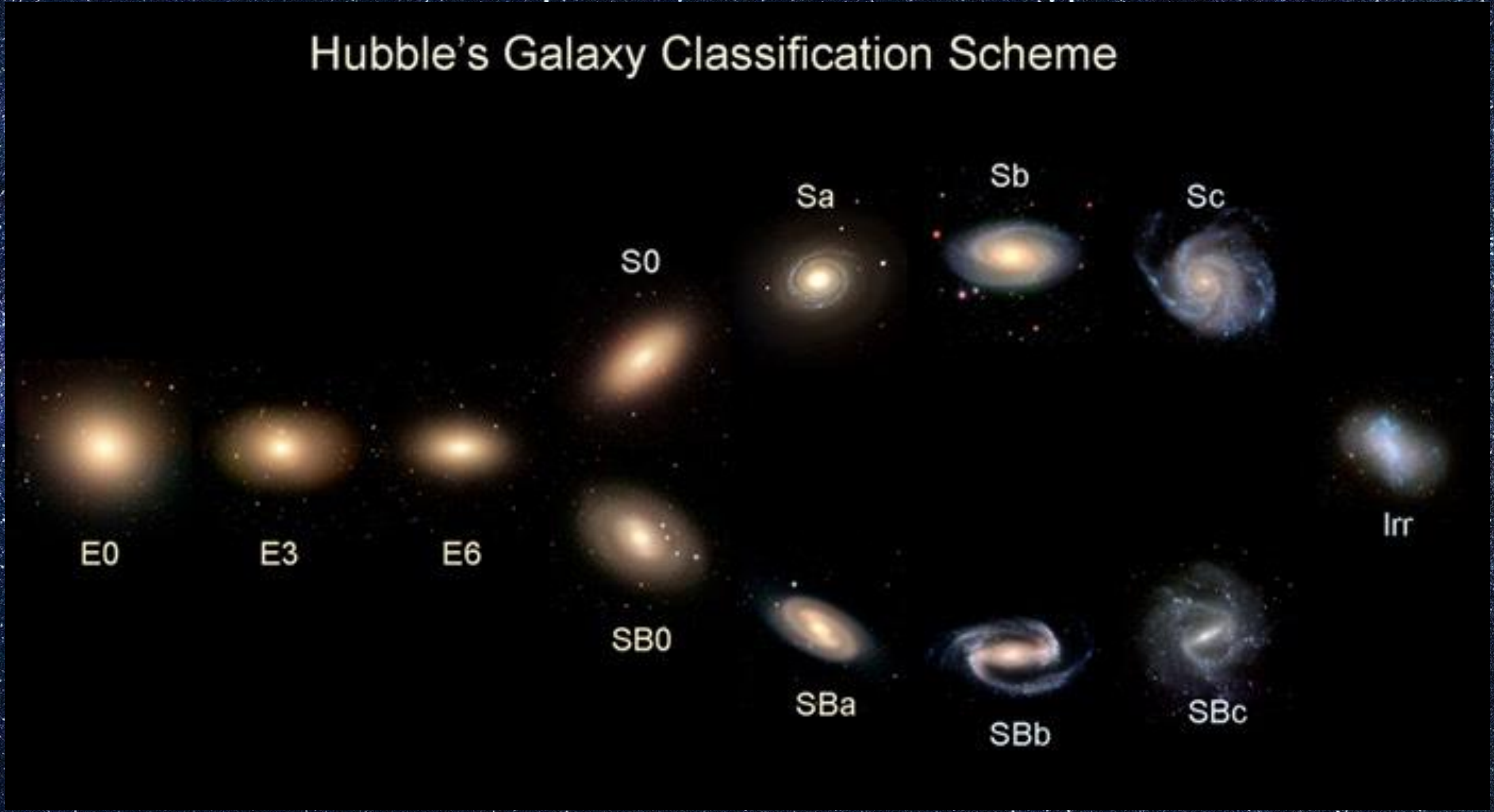
Intracluster Medium (ICM)

- Space between galaxies in a cluster is filled with hot, dense gas.
- Mass of this gas is greater than the combined (baryonic) mass of all the galaxies in the cluster
- Rich clusters shine more brightly
- Galaxies in the cluster crash into this gas a high speed

Credit: ESA/XMM-Newton/SDSS/J. Sanders et al. 2019

Types of galaxies
The Hubble tuning Fork

- Elliptical (E)
- Lenticular (S0)
- Spirals (S)
- Barred Spirals (SB)
- Irregulars (Irr)



Galaxy Zoo

- Dataset of 1 million galaxies from SDSS
- Want to understand more about physical processes that shape galaxy evolution
- Classify galaxies according to their morphology
- Too many for an individual us to!
- Citizen science project – ask the public
- Look through images and decide which type you think it is
- Initially just spiral, elliptical or merger

- https://www.zooniverse.org/projects/zookeeper/galaxy-zoo/

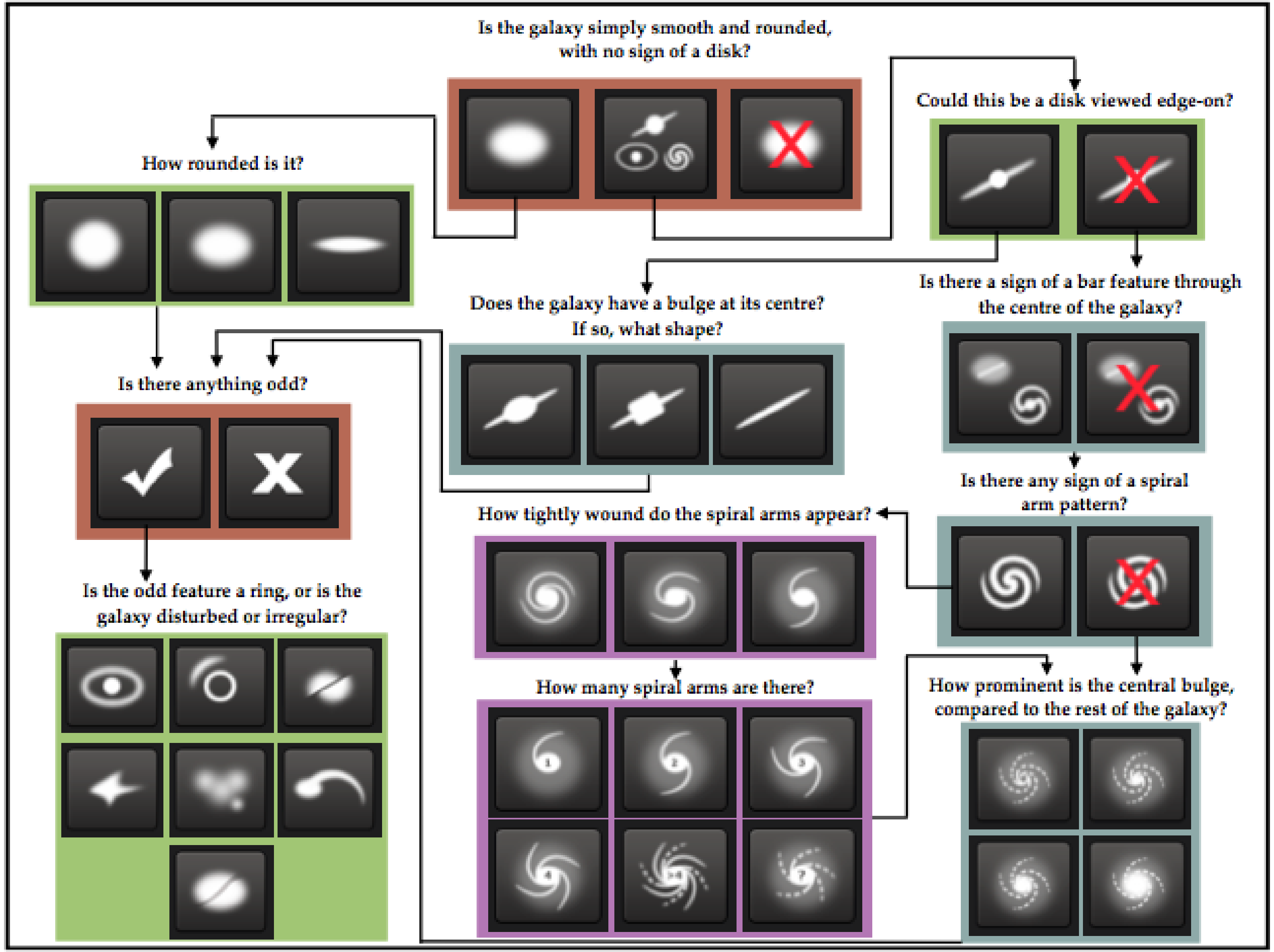


Figure 1. Flowchart of the classification tasks for GZ2, beginning at the top centre. Tasks are colour-coded by their relative depths in the decision tree. Tasks outlined in brown are asked of every galaxy. Tasks outlined in green, blue, and purple are (respectively) one, two or three steps below branching points in the decision tree. Table 2 describes the responses that correspond to the icons in this diagram.

Classification with Galaxy Zoo 2

- We will use the Galaxy Zoo 2 dataset to classify galaxies
- Labelled by citizen scientists
- As this is a very simple model, we will classify into two classes – spiral or elliptical.

Why use AI to study galaxies?

The problem:
Astronomers collect millions of galaxy images — too many to classify by eye.

The solution — Galaxy Zoo:
Volunteers label galaxies as spiral or elliptical, creating training data.

How AI helps:
AI learns from those labels and classifies new galaxies thousands of times faster.

Classification Pipeline

Galaxy Images

↓

Human Labels (Galaxy Zoo)

↓

AI Model (learns patterns)

↓

Galaxy Type Prediction ✓

What is a Neural Network?

A neural network is a computer model loosely inspired by the human brain.

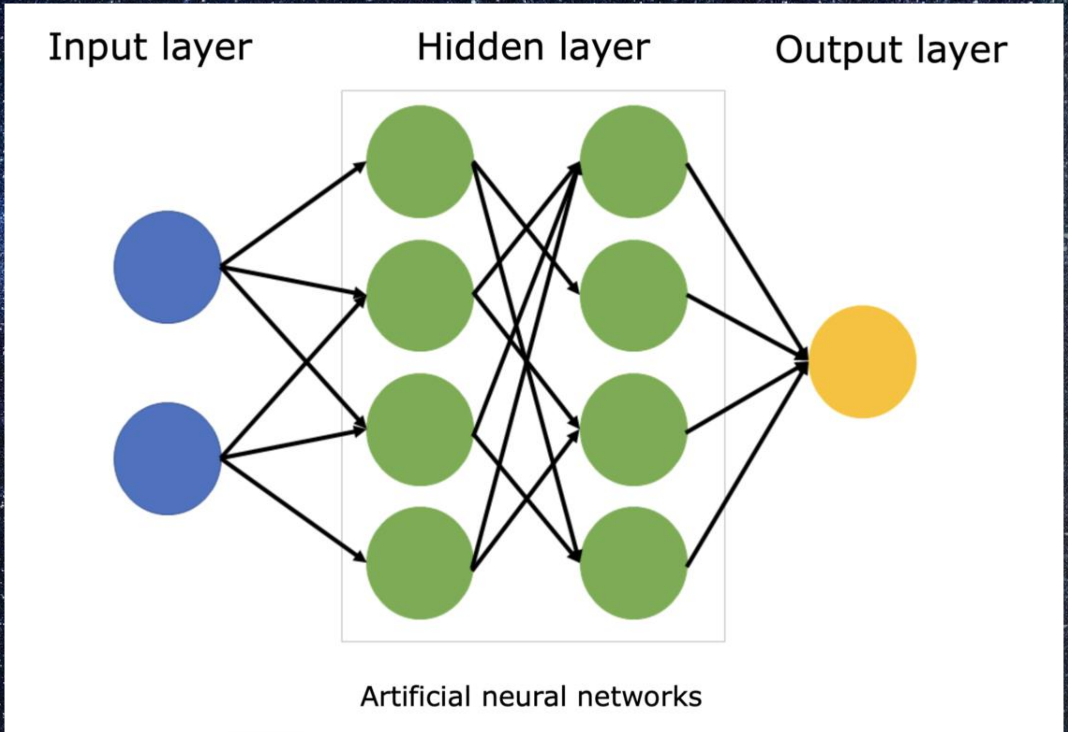
It is made of connected nodes (neurons) organised into layers:

• **Input layer —** receives the raw data (e.g. a galaxy image)

• **Hidden layers —** process and transform the data

• **Output layer —** gives the final answer

The model learns by studying many labelled examples and improving over time.



Neural Network Diagram

How does a neural network learn?

Weights

Each connection has a weight showing its importance.

Training

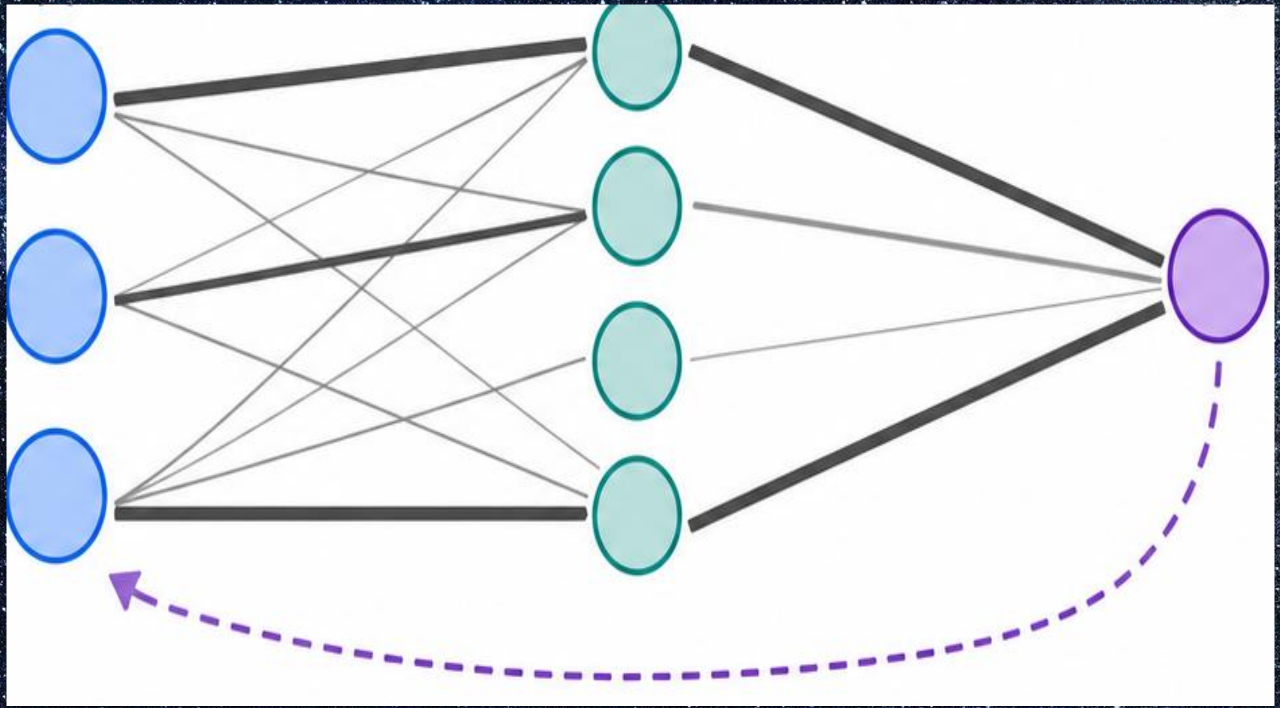
The model sees labelled examples and adjusts weights after each prediction.

Backpropagation

Errors flow backwards through the network, nudging every weight.

Result

After many examples, the model converges and makes accurate predictions.



What is a CNN? (Convolutional Neural Network)

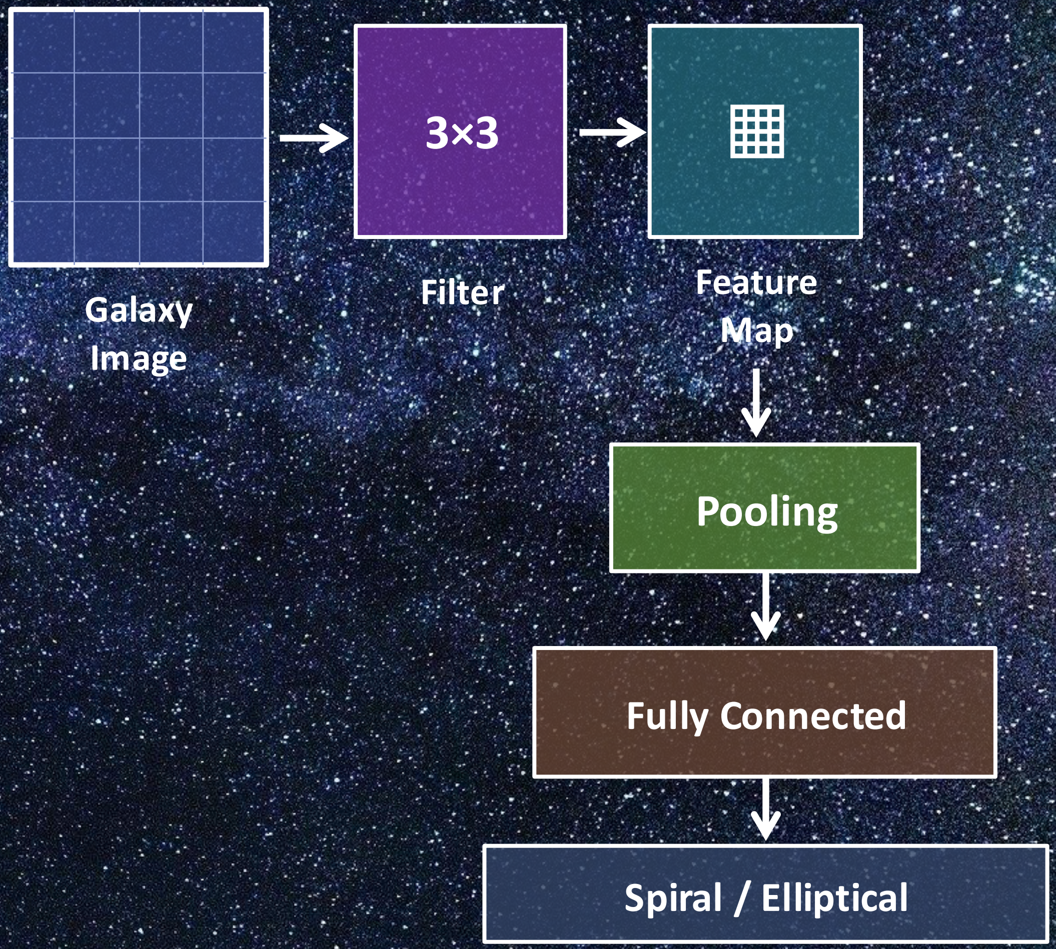
CNNs are specially designed for images.

Instead of looking at every pixel separately, a CNN scans the image with small filters to detect:

- Edges and boundaries
- Shapes and curves
- Bright regions / galaxy cores
- Spiral arms
- Smooth elliptical forms

The CNN learns these features automatically from the training data.

CNN Pipeline



Using CNNs to Classify Galaxies

Putting it all together:

1. Start with a large dataset of galaxy images
2. Pre-process images (resize, normalise)
3. Train the CNN on labelled examples
4. The CNN predicts the type of new, unseen galaxies

Why this matters:

Modern sky surveys produce hundreds of millions of images. AI lets astronomers analyse the whole dataset efficiently.

1 Galaxy Dataset
(millions of images)

↓

2 CNN Model
(learns features)

↓

3 Prediction
(galaxy type)

Output classes:

Spiral | Elliptical | Irregular

Pause to do the Classification task in python

Astrophysics of Galaxies: Ram Pressure Stripping

Ram Pressure Stripping

- Pressure exerted on an object moving through a fluid medium
- Space in-between galaxies in a cluster is filled with hot gas (ICM)
- Galaxies plunge into this gas at 1000s of km/s
- Gas is removed forming **long tails** behind the galaxy
- Signature is a **truncated gas disk**
  - Only strips gas, not stars!
- **Fast process** - can remove gas very quickly

Ram Pressure Stripping

- The ram pressure is given by:

$$P_{ram} = \rho u^2$$

- ρ – density of the ICM, u is the relative velocity of the galaxy and ICM
- Gas is removed when the force from ram pressure exceeds the gravitational force holding the gas inside the galaxy

$$P_{ram} \geq 2\pi G \Sigma(r)_g \Sigma(r)_{dg}$$

- Where Σ_g and Σ_{dg} are the surface densities of the gas and the gas plus the dark matter respectively.

Ram Pressure Stripping

- Gas is stripped from the outside-in
- Long tails of gas trailing behind the galaxy
- Can contain different phases of gas
  - Ionized, neutral, molecular
- Visible in multiple wavelengths

Cramer et al. 2019

Ram Pressure Stripping

- Should be more effective for **small galaxies in massive clusters**
  - Small galaxies have a smaller gravitational force holding the gas in
  - Massive clusters have larger densities and so stronger ram pressure
  - Galaxies travel faster in massive clusters

- Strength increases as galaxies move closer to the centre of the cluster

Stripped Galaxies

How do we know ram pressure stripping is responsible?

- Observational Evidence for RPS:
  - Truncated gas disk
  - Lower than expected gas mass
  - Stellar disk undisturbed
  - One-sided tails

What is happening to the gas?

Gas that is less tightly bound to the galaxy is removed

- Strip from the outside-in

Dense gas in molecular clouds is left behind

- More difficult to ‘push’ out

Gas remaining in the galaxy

- Is compressed rather than ‘pushed’
- Collapses to form more stars

Ram pressure is primarily responsible for removing gas

BUT

It can also increase star formation (only temporary)

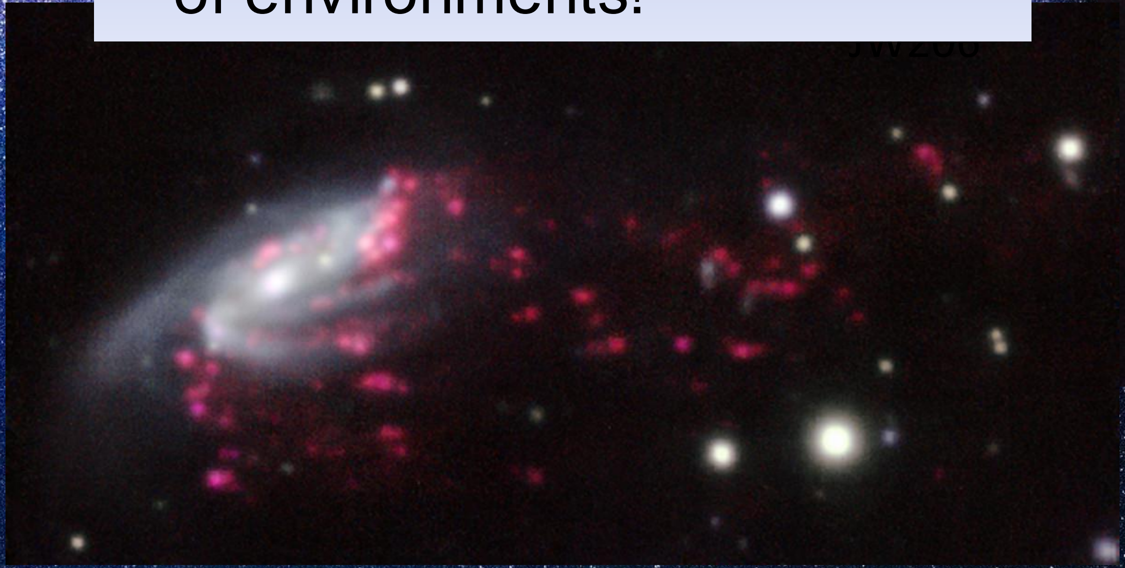
What is a Jellyfish galaxy?

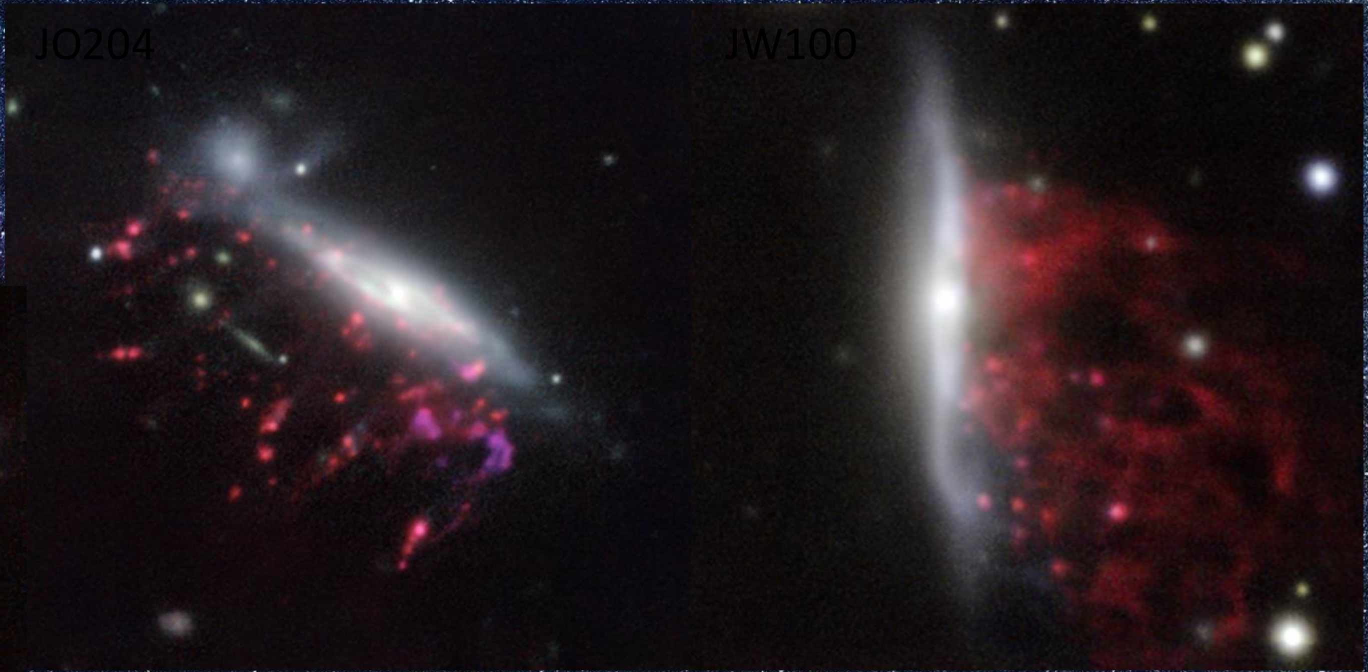
- Type of stripped galaxy
- Tails bright in multiple wavelengths
- Bright “tendrils” or “tentacles” of material due to ionized gas and/or star formation

Once rare objects but….

The GASP survey

- The GASP survey is discovering more and more jellyfish galaxies
- Range of galaxies in a range of environments!





The “tentacles” are ionized gas (red) lighter areas are star formation

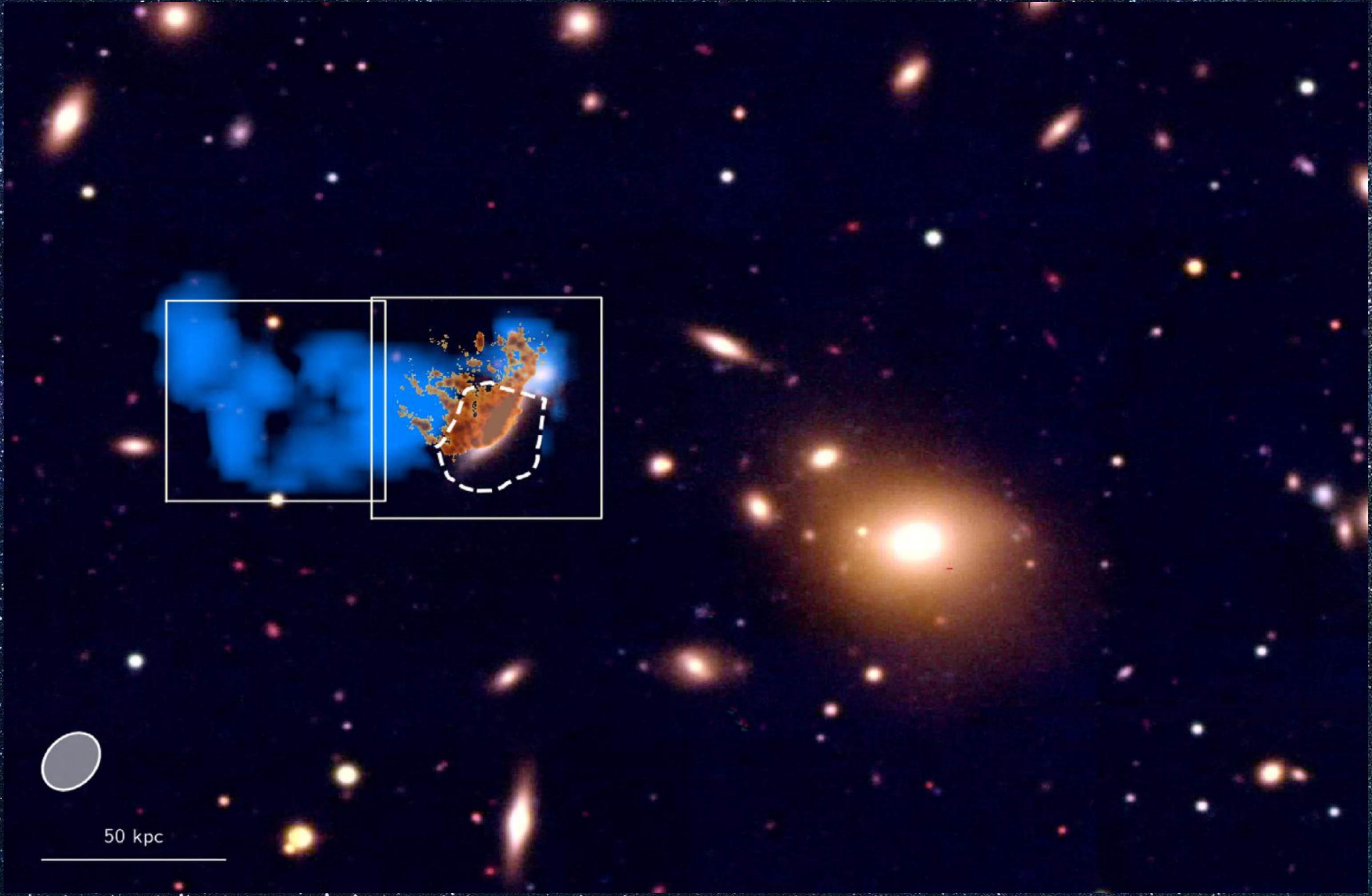
JO204

- Massive galaxy in a low mass cluster
- Red - glow from ionized hydrogen gas
- White regions are stars
- 40% of it’s gas mass has been stripped so far
- First infall to cluster centre

Credit:ESO/GASP collaboration

JO204

- Hα emission is orange (30kpc long)
- HI emission in blue (90kpc long)
- Asymmetric HI emission in galaxy – one side is stripped more than the other
- Molecular gas also detected in the tail



What is in a jellyfish tail?

- Diffuse (low density) Hα emitting gas
- Diffuse neutral (HI) gas
- Compact molecular clumps of gas
- Hot X-ray emitting gas
- Stars

We are finding more galaxies with this combined emission from the tail regions

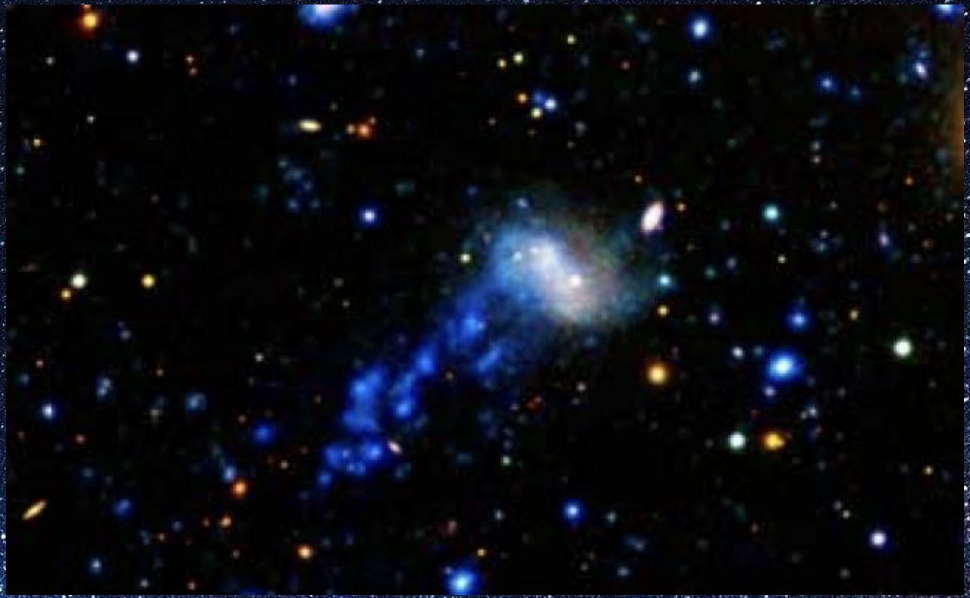
How are the different gas phases correlated?

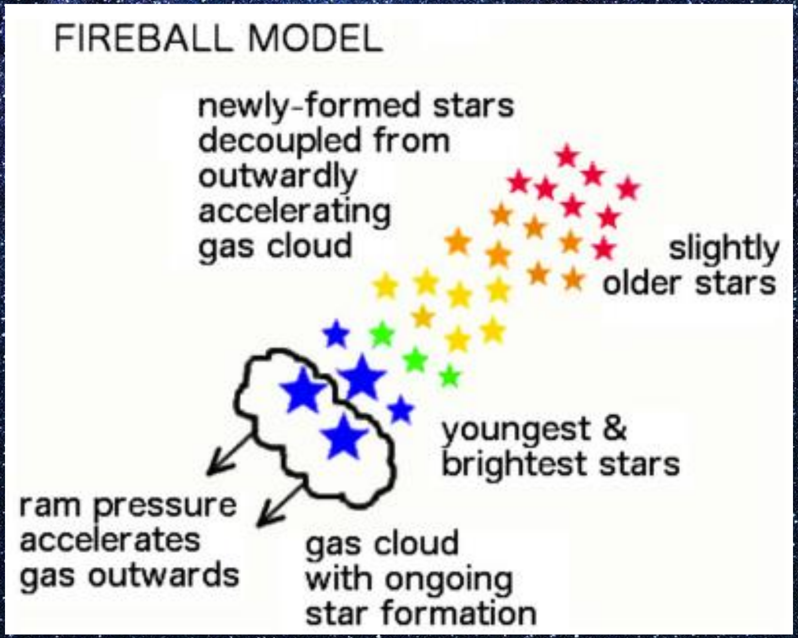
Different components are accelerated differently by ram pressure

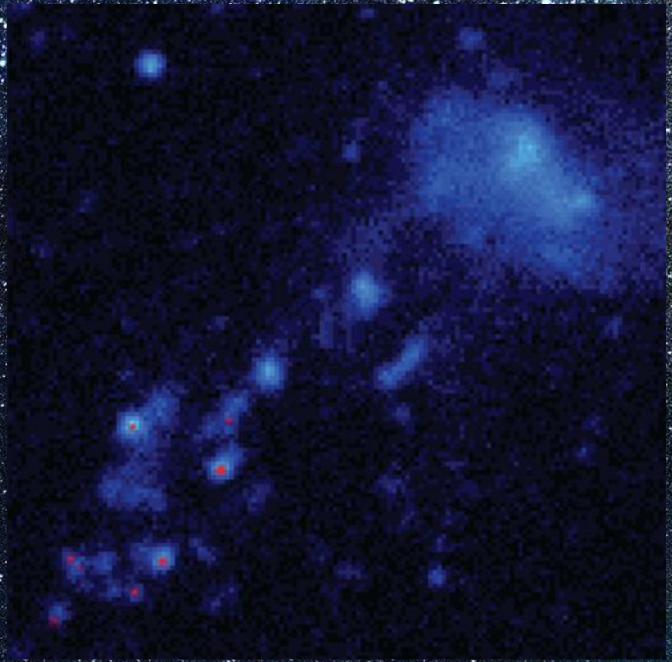
- Diffuse gas is easily pushed
- Dense clumps of gas are harder to push
- As the galaxy is stripped, these gas phases separate
- Stars are not affected by ram pressure
- Stars formed in the tail
  - Stay where they are
  - If close to the galaxy may fall back due to gravity

IC 3418

- Dwarf galaxy in the Virgo cluster
- Star formation has stopped
- UV bright tail (17kpc)
- No gas emission from galaxy
- Some neutral and ionized gas emission in tail
- Head-tail “fireball” features
- Hα emission in red offset from UV emission



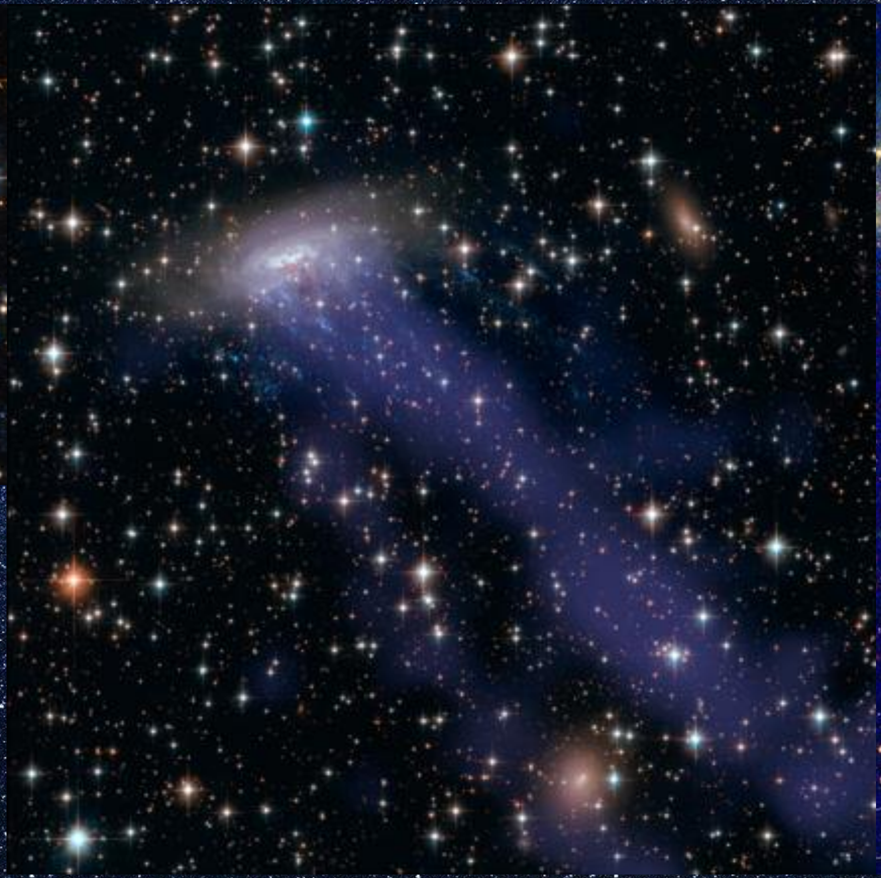




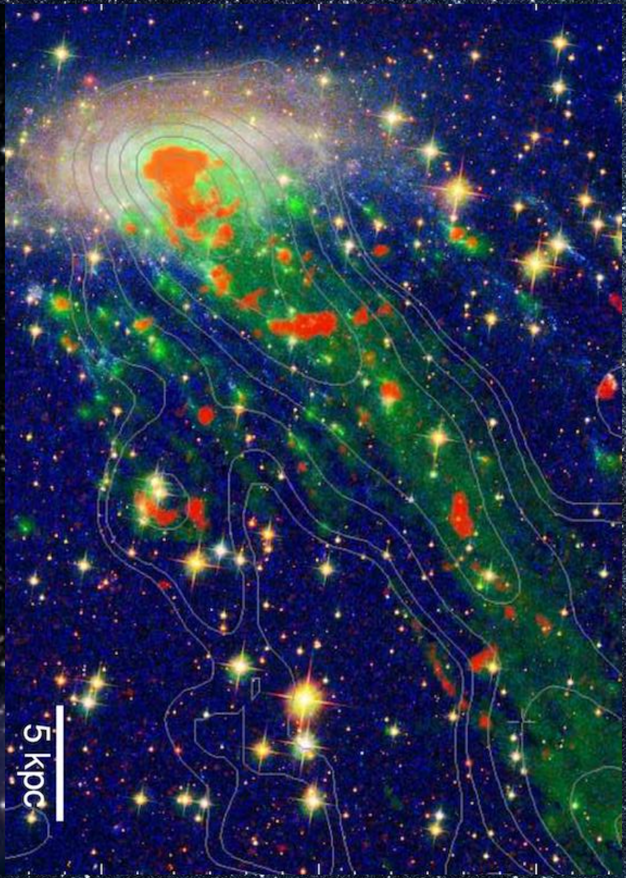
ESO 137-001

Optical + UV

X-ray

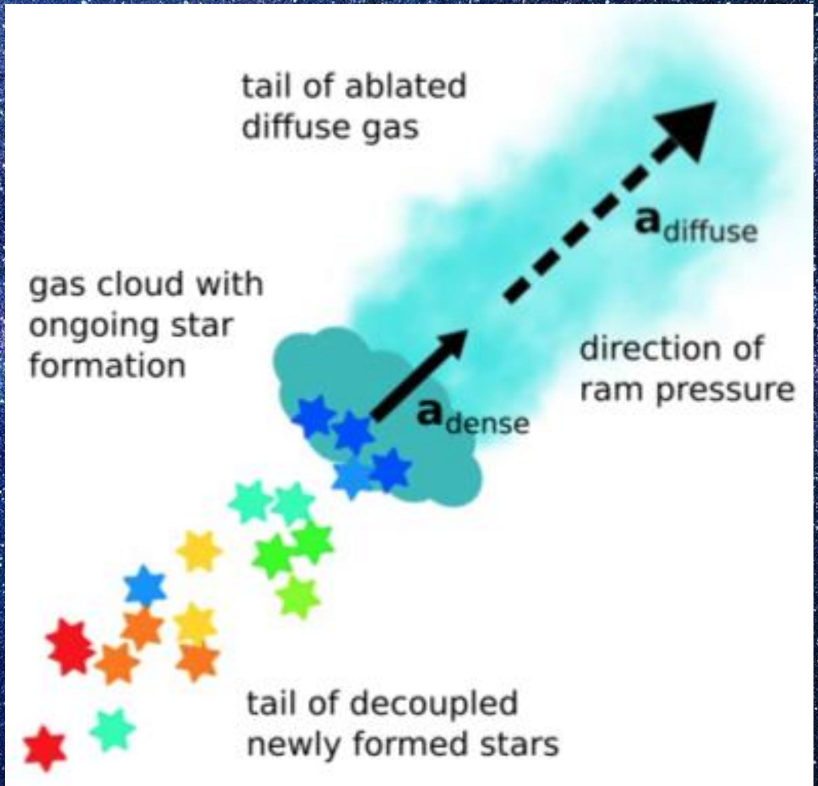


Hα CO X-ray



- Norma Cluster
- Travelling at 4.5 million miles an hour

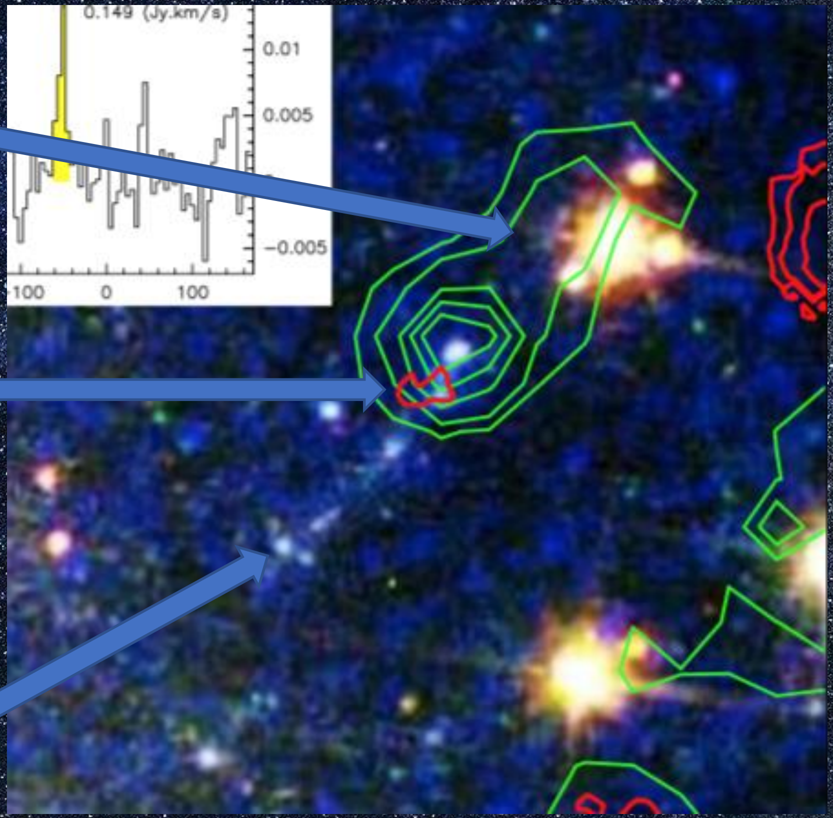
A Closer Look…



Low density Hα emitting gas
Easily accelerated by ram pressure

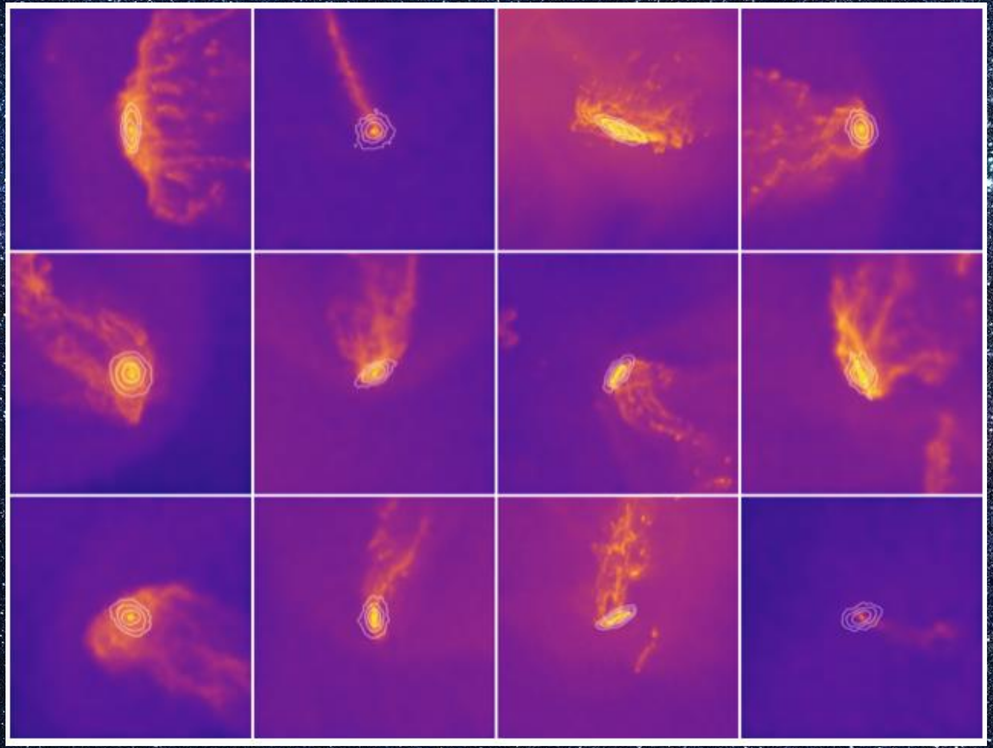
Central dense clump
Bright in Hα and/or CO
Harder to accelerate by ram pressure

Trail of stars
Not accelerated by ram pressure



Zooniverse Jellyfish Project

- You can help to classify jellyfish galaxies!
- Researches from the Max Plank Institute for Astronomy have simulated a virtual universe
- Illustris TNG simulations
- 100,000 satellite galaxies
- https://www.zooniverse.org/projects/apillepich/cosmological-jellyfish



Summary

- Internal processes such as supernova feedback have an impact on the evolution of a galaxy by:
  - Spreading metal rich gas around the galaxy
  - Ejecting gas from the galaxy
  - Blowing holes in the gas distribution to slow down the rate at which new stars form
- Most of the mass in the universe cannot be seen – dark matter!
- Evidence for this includes rotations curves and gravitational lensing.

Summary

- Galaxies morphologies can vary
- Their evolutionary path determines their morphology

- Galaxy clusters are harsh environments and processes such as ram pressure stripping can:
  - Strip the galaxy of gas, halting star formation
  - Change the morphology of the galaxy through changing is gravitational potential
  - Cause extragalactic star formation